

Sharp total variation bounds from spherical rearrangement

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Abstract

We study the sharp control of total variation by a higher derivative under homogeneous endpoint conditions. For every derivative order $n \geq 2$, we prove that the variation norm on $W_0^{n,2}(0,1)$ is twice the point-evaluation norm, with coincident midpoint-symmetric extremizers. A Jacobi-polynomial representation identifies the constrained Green form with a positive mixture of spherical Poisson interactions, permitting rearrangement to a cap. We also prove the complete measure-endpoint identity and equality classification in every order, using extreme moment measures, unimodal splines and contact motion. A weighted zero-count argument establishes strict midpoint height maximization and unique symmetric height extremizers at every finite exponent.

For each fixed order, compactness and persistence of a single peak give the full variation statement on intervals around two and immediately above one. We then prove finite nondegenerate switching and simple residual zeros for every finite-exponent variation maximizer, derive its switching Hessian, and exclude all two-switch and all midpoint-symmetric three-switch maximizers. An exact rational counterexample rules out an unrestricted extension to positive derivative weights.

Finally, we settle derivative order three for every finite exponent $1 < p < \infty$: the variation and height extremizers coincide, are symmetric about the midpoint, and have exactly one interior critical point. A conserved residual quantity parametrizes the nodal intervals by their quadratic vertices. A positive action is strictly concave in the vertex depths and strictly convex after optimization in the mean parameter. Comparing its values for different nodal counts constructs a better competitor whenever there are at least four intervals; the earlier two-switch exclusion completes the proof. At the uniform endpoint, a duality and regularity argument extends the positive action and gives the complete equality classification with an explicit extremizer. Thus the order-three conjecture holds at every exponent, with the measure interpretation at one. In every order, an endpoint conserved quantity also proves saturation of the highest derivative, finitely many simple critical points and simple residual zeros. For each fixed order, the critical-point count is uniformly bounded over all exponents above one, including the uniform endpoint. The endpoint switching objective has an explicit Hessian supported on the simple residual zeros. A semialgebraic description of the resulting finite splines proves that the uniform-endpoint sharp variation constant is real algebraic in every order, with an algebraic maximizing spline. The full all-order variation conjecture remains open for the remaining cases in orders at least four.

1. The sharp inequality

All functions and measures in this paper are real. For $1 \leq p \leq \infty$, we use the homogeneous norm $\|u^{(n)}\|_p$ on $W_0^{n,p}(0,1)$. The endpoint conditions are

$$u^{(j)}(0) = u^{(j)}(1) = 0, \quad 0 \leq j < n. \quad (1)$$

Let $C_{n,p}$ and $V_{n,p}$ denote the sharp constants in

$$\|u\|_\infty \leq C_{n,p} \|u^{(n)}\|_p, \quad \|u'\|_1 \leq V_{n,p} \|u^{(n)}\|_p. \quad (2)$$

The inequality $V_{n,p} \geq 2C_{n,p}$ follows from the variation of a function that vanishes at both endpoints. Nazarov and Shcheglova [NS, Conjecture 4.14] conjecture equality for all $n \geq 2$ and $1 \leq p \leq \infty$, together with coincidence and midpoint symmetry of the extremizers. Their Remark 4.15 records the previously known case $n = 2$.

Theorem 1. For every integer $n \geq 2$,

$$V_{n,2} = 2C_{n,2} = \frac{1}{2^{2n-2}(n-1)!\sqrt{2n-1}}. \quad (3)$$

The nonzero extremizers are precisely the scalar multiples of the Riesz representer $R_n(\cdot, 1/2)$ of evaluation at $1/2$ in $W_0^{n,2}(0,1)$. In particular, the maximizing sets in (2) coincide, and their elements are even about $1/2$.

The point-evaluation formula used here is prior work. Kalyabin [K, Theorem 2] gives, after affine rescaling,

$$\|\text{ev}_a\|^2 = \frac{[a(1-a)]^{2n-1}}{(2n-1)((n-1)!)^2}. \quad (4)$$

Our contribution is the upper bound for the variation norm and its equality classification, obtained through the rearrangement principle in Section 3. Best polynomial approximation already occurs in this embedding problem [GS]. The sharp $W_0^{m,2}$ to L^1 embedding of the function itself was proved by Hindov, Nitzan, Olsen and Rydhe [HNOR]; their theorem does not include the additional zero-mean constraint on a derivative that appears below.

2. A Jacobi representation of the clamped Green form

Fix $m \geq 1$, and set

$$w(x) = x^m(1-x)^m, \quad M = \int_0^1 w(x) dx = \frac{(m!)^2}{(2m+1)!}, \quad d\pi(x) = M^{-1}w(x) dx. \quad (5)$$

Let h_k be the orthonormal polynomials for π , with positive leading coefficients. These are normalized shifted Jacobi polynomials with parameters (m, m) ; in particular $h_0 = 1$. Put

$$v_k = wh_k, \quad \lambda_k = \frac{(k+2m)!}{k!}. \quad (6)$$

Lemma 2. The functions $v_k/\sqrt{M\lambda_k}$ form an orthonormal basis of $H = W_0^{m,2}(0,1)$ in its energy inner product. Moreover,

$$(-1)^m v_k^{(2m)} = \lambda_k h_k. \quad (7)$$

Proof. The left side of (7) has degree k . If q is a polynomial of degree less than k , integration by parts $2m$ times yields

$$\int_0^1 (wh_k)^{(2m)} wq = \int_0^1 wh_k (wq)^{(2m)} = 0. \quad (8)$$

Every boundary term vanishes because one of its two factors has fewer than m derivatives and retains an endpoint zero. Orthogonality therefore shows that the left side of (7) is proportional to h_k . Comparing leading coefficients proves (7). A further integration by parts gives

$$\langle v_k, v_\ell \rangle_H = M\lambda_k \delta_{k\ell}. \quad (9)$$

If $y \in H$ is orthogonal to every v_k , weak integration by parts gives $\int_0^1 y h_k = 0$ for every k . The h_k span the polynomials, which are dense in $C[0,1]$, so $y = 0$. This proves completeness. $\square$

The mean functional $y \mapsto \int_0^1 y$ has Riesz representer $v_0/\lambda_0 = w/(2m)!$. Therefore

$$H_* := \left\{ y \in H : \int_0^1 y = 0 \right\} \quad (10)$$

is obtained by removing the zeroth basis vector. For a bounded real load g , Parseval gives the squared dual norm of $y \mapsto \int gy$ on H_* :

$$Q_m(g) = M \sum_{k \geq 1} \frac{\left| \int_0^1 g h_k d\pi \right|^2}{\lambda_k}. \quad (11)$$

Equivalently, if G_m is the clamped Green kernel and $w_m = w/(2m)!$, then

$$Q_m(g) = \iint g(x) G_m(x, t) g(t) dx dt - \frac{(\int g w_m)^2}{\int w_m}. \quad (12)$$

Formula (11) is sufficient for the proof and avoids any pointwise convergence issue for a Green series.

3. Spherical rearrangement of the load

Set $d = 2m + 2$ and let σ be normalized surface measure on S^d . The coordinate $x = (1 + \xi_{d+1})/2$ has law π . Thus

$$F(\xi) = g\left(\frac{1 + \xi_{d+1}}{2}\right) \quad (13)$$

identifies $L^2(\pi)$ with the zonal subspace of $L^2(S^d)$. Under this identification, h_k is the normalized zonal harmonic of degree k . This follows from the orthogonal decomposition of polynomials into

spherical harmonics: a zonal polynomial of degree k orthogonal to all lower zonal degrees has only its degree- k component.

The spherical Poisson operator P_r , $0 < r < 1$, has eigenvalue r^k on degree- k harmonics and kernel

$$K_r(\xi \cdot \eta) = \frac{1 - r^2}{(1 - 2r\xi \cdot \eta + r^2)^{(d+1)/2}}. \quad (14)$$

We use normalized surface measure, as in [ABR, (1.15)]. The eigenvalue follows by harmonic extension of a homogeneous harmonic polynomial. The beta integral gives

$$\frac{1}{\lambda_k} = \frac{1}{(2m-1)!} \int_0^1 r^k (1-r)^{2m-1} dr. \quad (15)$$

Tonelli's theorem applied to the nonnegative spectral terms in (11) therefore gives

$$Q_m(g) = \frac{M}{(2m-1)!} \int_0^1 (1-r)^{2m-1} \left(\langle F, P_r F \rangle - \left(\int F d\sigma \right)^2 \right) dr. \quad (16)$$

The deletion of the constant harmonic is exactly the mean subtraction needed in (12).

Theorem 3. If $A \subset (0, 1)$ is measurable and $a \in [0, 1]$ satisfies $\pi(A) = \pi([0, a])$, then

$$Q_m(\mathbf{1}_A) \leq Q_m(\mathbf{1}_{[0,a]}). \quad (17)$$

If $0 < \pi(A) < 1$, equality holds only when A agrees, up to null sets, with an interval abutting one endpoint and having that π -measure.

Proof. Lift A to the zonal subset $E \subset S^d$ using (13). For fixed r , K_r is positive, bounded and strictly decreasing in geodesic distance. Spherical rearrangement [BH, Theorem 2] maximizes the self-interaction of $\mathbf{1}_E$ at fixed surface measure by a cap. Choose that cap about the negative coordinate axis. It corresponds to $[0, a]$. The mean term in (16) is unchanged, and integration proves (17).

For clarity, the equality hypotheses in [BH] are satisfied with two functions and integrand $(s, t) \mapsto st$. Its mixed finite difference is strictly positive; the kernel is positive and strictly decreasing; and the nontrivial indicators are nonconstant with finite interaction. Hence equality at fixed r forces E to be a cap up to rotation. If equality holds in (17), the nonnegative rearrangement deficit has zero integral in (16), so equality holds for almost every r , and in particular for one $r \in (0, 1)$. Thus E is a cap.

Its axis must be the original coordinate axis up to reversal. Indeed, zonal invariance makes the centroid of E parallel to that axis. A nontrivial cap has a nonzero centroid parallel to its own axis. Consequently A is an endpoint interval, as claimed. $\square$

The same argument proves a broader load rearrangement statement. For a bounded real g , let $g^\downarrow$ be its nonincreasing equimeasurable rearrangement relative to π . Adding a constant makes g nonnegative without changing Q_m . Applying [BH] in (16) gives

$$Q_m(g) \leq Q_m(g^\downarrow). \quad (18)$$

This is rearrangement of a load in a Green quadratic form. It does not assert that ordinary rearrangement preserves a higher derivative norm.

4. Proof of Theorem 1

Put $m = n - 1$. The map $u \mapsto y = u'$ is an isometric bijection from $W_0^{n,2}$ to H_* : integrating y recovers u , and the zero-mean condition supplies $u(1) = 0$.

Duality gives

$$V_{n,2}^2 = \sup_{|g| \leq 1} Q_m(g) = 4 \sup_A Q_m(\mathbf{1}_A). \quad (19)$$

To justify the second equality, write a bounded g with $|g| \leq 1$ as the average of $\text{sign}(g - s)$ over uniform $s \in [-1, 1]$. Convexity of the squared Hilbert dual norm bounds its value by the supremum over sign functions. Constants are annihilated in (11), giving $Q_m(2\mathbf{1}_A - 1) = 4Q_m(\mathbf{1}_A)$.

For an endpoint interval,

$$\int_0^a y(x) dx = u(a), \quad Q_m(\mathbf{1}_{[0,a]}) = \|\text{ev}_a\|^2. \quad (20)$$

Theorem 3, (4), and (19) now prove (3). In particular, the maximizing interval is a half-interval.

Suppose that $u \neq 0$ attains equality, and take $g = \text{sign}(u')$, choosing either sign where $u' = 0$. Both signs occur on sets of positive measure. Equality in the Hilbert dual bound and in Theorem 3 forces g to be a half-interval sign function, up to its global sign and null sets. Its functional on H_* is $\pm 2u(1/2)$. Equality in Cauchy--Schwarz therefore makes u a scalar multiple of $R_n(\cdot, 1/2)$.

Conversely, this representer maximizes point evaluation. Its variation is at least twice its maximum and at most the bound just proved, so it also maximizes variation. Formula (4) has its unique maximum at $a = 1/2$, and uniqueness of a Hilbert-space Riesz representer gives precisely the same classification for the point-norm extremizers. Finally, reflection preserves the energy inner product and fixes evaluation at $1/2$, so uniqueness makes $R_n(\cdot, 1/2)$ reflection invariant. $\square$

5. The measure endpoint

Let $\mathcal{M}_n$ consist of real finite signed measures μ on $[0, 1]$ such that

$$\|\mu\|_{\text{TV}} \leq 1, \quad \int_0^1 t^j d\mu(t) = 0 \quad (0 \leq j < n). \quad (21)$$

Their compactly supported potentials are

$$U_\mu(x) = \int_0^1 \frac{(x-t)_+^{n-1}}{(n-1)!} d\mu(t). \quad (22)$$

The moments make U_μ vanish off $[0, 1]$, and $D^n U_\mu = \mu$. For $n \geq 2$, U_μ is continuous and absolutely continuous, with integrable first derivative. Endpoint atoms are allowed in this relaxed formulation.

Theorem 4. For every $n \geq 2$,

$$\sup_{\mu \in \mathcal{M}_n} \|U'_\mu\|_1 = 2 \sup_{\mu \in \mathcal{M}_n} \|U_\mu\|_\infty. \quad (23)$$

The maximizing measures for the two quantities coincide. The relaxed constants equal $V_{n,1}$ and $C_{n,1}$, respectively. No symmetry assertion at $p = 1$ is included.

Proof of the constant identity. Every extreme point of $\mathcal{M}_n$ has norm one and exactly $n + 1$ support points. To prove the support bound, suppose μ has norm one and at least $n + 2$ disjoint Borel sets of positive $|\mu|$ -measure. A nonzero function h constant on those sets can satisfy the n equations $\int t^j h d\mu = 0$ and the extra equation $\int h d|\mu| = 0$. Then $(1 \pm \varepsilon h)\mu$ are distinct norm-one admissible measures for small ε . Such a μ is not extreme. A nonzero measure of norm less than one is excluded by scalar perturbation. Zero is the midpoint of opposite nonzero moment measures, which exist on any $n + 1$ distinct nodes, and is not extreme either. A nonzero measure on at most n points is excluded by the Vandermonde determinant.

On nodes $t_0 < \dots < t_n$, the moment nullspace is one-dimensional, with weights

$$b_i = \prod_{j \neq i} (t_i - t_j)^{-1}. \quad (24)$$

Thus every extreme measure is a normalized signed multiple of $\sum_i b_i \delta_{t_i}$. Its potential is a B-spline, up to scale and sign. More explicitly, the Genocchi--Hermite identity [dB, (52)] identifies

$$\rho(x) = n \sum_{i=0}^n b_i (t_i - x)_+^{n-1} = (-1)^n n! U_{\sum b_i \delta_{t_i}}(x) \quad (25)$$

with the density of $\sum_i t_i X_i$, where X is uniform on the standard n -simplex. After parallel sections of the simplex are identified by translation, their Minkowski interpolation is contained in the section at the interpolated coordinate. The classical Brunn--Minkowski inequality [GHW, (3), $p = 1$] therefore makes their volumes to the power $1/(n - 1)$ concave. The coarea factor relating section volume to ρ is constant, so $\rho^{1/(n-1)}$ is concave and ρ is unimodal. It is positive between its end knots, continuous, and zero at those knots. Every extreme potential therefore satisfies

$$\|U'_\mu\|_1 = 2\|U_\mu\|_\infty. \quad (26)$$

Write $C = \sup_{\mathcal{M}_n} \|U_\mu\|_\infty$. The set $\mathcal{M}_n$ is weak-star compact. Moreover, $\mu \mapsto \|U'_\mu\|_1$ is weak-star lower semicontinuous: it is the supremum of the continuous linear functionals $\int H_g d\mu$, where $|g| \leq 1$ and

$$H_g(t) = \int_t^1 \frac{(x - t)^{n-2}}{(n - 2)!} g(x) dx. \quad (27)$$

Its sublevel set at $2C$ is closed and convex and contains all extreme points by (26). Krein--Milman therefore gives the upper bound in (23). The elementary variation lower bound gives the reverse inequality. The height supremum is attained: on the bounded measure set, joint continuity of the kernels in (22) makes the potential map weak-star to uniform continuous. Consequently the variation supremum is attained as well.

Coincidence. Let μ maximize variation, choose $g = \text{sign}(U'_\mu)$, and let $\mathcal{F}$ be the compact exposed face on which $\int H_g d\nu = 2C$. Its extreme points are extreme in $\mathcal{M}_n$. For each such point e ,

$$2C = \int g U'_e \leq \|U'_e\|_1 = 2\|U_e\|_\infty \leq 2C. \quad (28)$$

Thus g agrees with the sign of U'_e wherever the latter is nonzero, and every U_e maximizes height. Its end knots must be 0 and 1: a potential supported on a shorter interval of length L can be stretched to $[0, 1]$, keeping its derivative-measure norm one and multiplying its height by $L^{1-n} > 1$.

All U_e have the same global sign, since opposite signs give opposite nonzero derivatives on a common interval near zero, contradicting their agreement with g . Change that common sign to positive. Concavity of $U_e^{1/(n-1)}$ makes its maximum set a closed interval, with strictly positive derivative almost everywhere before it and strictly negative derivative almost everywhere after it. Two disjoint maximum intervals would again force contradictory signs for g . The maximum intervals therefore intersect pairwise, and hence have a common point a_* . All $U_e(a_*) = C$. Evaluation is weak-star continuous, so Krein--Milman applied to $\mathcal{F}$ gives $U_\mu(a_*) = C$. Conversely, a height maximizer is a variation maximizer by the already proved constant identity.

Recovery of the ordinary constants. For an ordinary admissible u , the measure $u^{(n)}dx$ satisfies (21). Conversely, mollify the zero extension of U_μ with a nonnegative mollifier supported in $[-\varepsilon, \varepsilon]$, obtaining U_ε . With $a = 1 + 2\varepsilon$, set

$$u_\varepsilon(x) = a^{1-n}U_\varepsilon(ax - \varepsilon). \quad (29)$$

It is smooth, supported on $[0, 1]$, satisfies all endpoint conditions, and has $\|u_\varepsilon^{(n)}\|_1 \leq 1$. Its maximum and variation converge to those of U_μ , by uniform convergence of the potential mollification and L^1 convergence of its first derivative. This proves equality of the relaxed and ordinary sharp constants. The proof classifies coincidence of relaxed maximizers and does not assert attainment in the ordinary $W_0^{n,1}$ class. $\square$

6. An open interval of exponents

The Hilbert theorem also yields a neighborhood of exponents for each fixed derivative order. The interval is not asserted to be uniform in the order. In this section write $X_{n,p} = W_0^{n,p}(0, 1)$ for the clamped space defined by (1), with norm $\|u^{(n)}\|_p$.

Theorem 5 (local continuation in the exponent). For every fixed integer $n \geq 3$, there exists $0 < \varepsilon_n < 1$ such that, if $|p - 2| < \varepsilon_n$, then

$$V_{n,p} = 2C_{n,p}. \quad (6.1)$$

All nonzero extremizers in the two inequalities are the scalar multiples of the unique positive norm-one extremizer for evaluation at $1/2$. They are symmetric about the midpoint and, after a choice of global sign, have strictly positive derivative on $(0, 1/2)$ and strictly negative derivative on $(1/2, 1)$.

The radius is allowed to depend on n and is not estimated here. The theorem does not reach all $1 < p < \infty$, give an interval uniform in n , or settle the full all-exponent variation conjecture.

6.1. Dual representation and compactness of variation maximizers

Let $\mathcal{P}_{n-1}$ denote the polynomials of degree at most $n - 1$. Integration from the left endpoint identifies $X_{n,p}$ isometrically with

$$Z_{n,p} = \{f \in L^p(0, 1) : \int_0^1 t^j f(t) dt = 0, \ 0 \leq j < n\}. \quad (6.2)$$

Indeed $u(x) = \int_0^x (x - t)^{n-1} f(t) / (n - 1)! dt$; the right endpoint conditions are equivalent to the displayed moments. For bounded g , put

$$H_g(t) = \int_t^1 \frac{(x-t)^{n-2}}{(n-2)!} g(x) dx. \quad (6.3)$$

Fubini gives $\int g u' = \int H_g f$. If $q = p/(p-1)$, the norm of this functional on $Z_{n,p}$ equals

$$\min_{P \in \mathcal{P}_{n-1}} \|H_g - P\|_q. \quad (6.4)$$

Here is a direct justification without a quotient-duality assumption. A minimizing polynomial exists by finite-dimensional coercivity and is unique for $q > 1$. Its residual $R = H_g - P$ satisfies

$$\int_0^1 |R|^{q-2} R Q = 0 \quad (Q \in \mathcal{P}_{n-1}). \quad (6.5)$$

If $R \neq 0$, the function $f = \text{sign}(R)|R|/\|R\|_q^{q-1}$ has L^p norm one and belongs to (6.2). It realizes the bound in Hölder's inequality. The zero-residual case is immediate. This proves (6.4).

For every fixed $p > 1$, the variation supremum is attained. A norm-bounded sequence is bounded in $W^{n,p}$ by successive integration, and its derivatives through order $n-1$ are uniformly bounded and equicontinuous, using Hölder's inequality for the highest integral. A subsequence converges uniformly through order $n-1$, and the highest derivatives have a weakly convergent subsequence in L^p . The limit satisfies all endpoint conditions, has norm at most one, and retains the limiting variation. The supremum is positive, so a maximizer must have norm exactly one by scaling.

For a norm-one variation maximizer u , choose $g = \text{sign}(u')$, with any value in $[-1, 1]$ on the zero set. The functional in (6.4) is at most $V_{n,p}$, since $\int g v' \leq \|v'\|_1$. On u it equals $V_{n,p}$. Thus its residual satisfies $\|R\|_q = V_{n,p}$, and equality in Hölder gives

$$u^{(n)}(t) = \text{sign } R(t) \left| \frac{R(t)}{V_{n,p}} \right|^{q-1}. \quad (6.6)$$

This formula gives more compactness than the original Sobolev bound.

Lemma 6 (compactness of maximizers). Suppose $p_j \rightarrow 2$ and u_j is any sequence of norm-one variation maximizers in X_{n,p_j} . Then, after a subsequence and a choice of global signs, u_j converges in $C^n[0, 1]$ to the positive norm-one Hilbert maximizer u_* .

Proof. Restrict to $3/2 \leq p_j \leq 5/2$, so $5/3 \leq q_j \leq 3$. The functions H_{g_j} , with $|g_j| \leq 1$, are uniformly bounded and Lipschitz. For $n = 2$, this follows directly from $H'_g = -g$; for $n \geq 3$, differentiate (6.3) once and bound its remaining integral.

The minimizing polynomials satisfy $\|P_j\|_{q_j} \leq 2\|H_{g_j}\|_{q_j}$. Their coefficients are uniformly bounded. To verify the needed uniform norm equivalence, a contrary sequence of polynomials can be normalized to have coefficient norm one and L^{q_j} norm tending to zero. A coefficient-convergent subsequence would give a nonzero polynomial vanishing almost everywhere, a contradiction. Consequently both P_j and their derivatives are uniformly bounded, and the residuals R_j are uniformly bounded and Lipschitz.

The quantities V_{n,p_j} are bounded above by (6.4) with the zero polynomial. They are bounded away from zero by testing the one fixed nonzero function u_* , whose highest derivative is bounded. Formula (6.6) now makes $u_j^{(n)}$ uniformly bounded and equicontinuous. For example, signed power maps with exponent in $[2/3, 2]$ on a fixed bounded interval have a common Hölder modulus with exponent $2/3$, after enlarging the constant. Arzelà--Ascoli supplies a uniformly convergent subsequence of the highest derivatives. Integration and the left endpoint conditions give convergence in C^n to some u .

Uniform boundedness and $p_j \rightarrow 2$ imply $\int |u^{(n)}|^2 = \lim_j \int |u_j^{(n)}|^{p_j} = 1$. All right endpoint conditions pass to the limit. Also $\|u'\|_1 = \lim_j V_{n,p_j}$. Testing $u_*/\|u_*^{(n)}\|_{p_j}$ gives $\liminf_j V_{n,p_j} \geq V_{n,2}$, while the Hilbert inequality applied to the limit gives the reverse bound. Thus u is a Hilbert maximizer. Theorem 1 identifies it with u_* or $-u_*$, proving the lemma. $\square$

6.2. Persistence of a single peak

We give the details that weak or uniform convergence alone would miss. The Hilbert midpoint representer, up to a positive factor, can be written

$$U(x) = F(|1 - 2x|), \quad F(z) = \int_z^1 (t^2 - z^2)^{n-1} dt, \quad 0 \leq z \leq 1. \quad (6.7)$$

To check this representation, expand the integrand. The resulting expression for F is an even polynomial of degree at most $2n - 2$, plus a multiple of z^{2n-1} . Thus (6.7) is polynomial of degree at most $2n - 1$ on each half-interval and has matching derivatives through order $2n - 2$ at the midpoint. Near $z = 1$, writing $t = z + (1 - z)s$ shows that $F(z)$ is a positive constant times $(1 - z)^n$, to leading order. Therefore all endpoint derivatives below order n vanish. The jump of the derivative of order $2n - 1$ is nonzero. The weak derivative $(-1)^n D^{2n} U$ is a scalar multiple of the midpoint delta. Testing against U shows this scalar is positive because $\int |U^{(n)}|^2 > 0$ and $U(1/2) > 0$. Uniqueness of the Riesz representer proves the claim up to positive normalization.

For $n \geq 2$, differentiation gives

$$F'(z) = -2(n - 1)z \int_z^1 (t^2 - z^2)^{n-2} dt < 0 \quad (0 < z < 1). \quad (6.8)$$

Also $F''(0) = -2(n - 1)/(2n - 3) < 0$. Hence $u'_* > 0$ on $(0, 1/2)$, $u'_* < 0$ on $(1/2, 1)$, and $u''_*(1/2) < 0$. The endpoint expansion gives $u_*^{(n)}(0) > 0$ and $(-1)^n u_*^{(n)}(1) > 0$.

These properties imply that every sufficiently close C^n perturbation satisfying the same clamped endpoint conditions has exactly one interior critical point, a strict maximum. Indeed, near zero its highest derivative remains positive and

$$u'(x) = \int_0^x \frac{(x - t)^{n-2}}{(n - 2)!} u^{(n)}(t) dt > 0. \quad (6.9)$$

The reflected argument gives negativity near one. On compact intervals away from the endpoints and midpoint the strict derivative signs persist. On a small interval about the midpoint, the second derivative stays negative, so there is precisely one zero of the first derivative, bracketed by its signs on either side.

Lemma 6 therefore implies that for all p sufficiently close to two, every norm-one variation maximizer is, up to sign, positive and strictly unimodal. Such a maximizer has variation exactly twice its height. Since every admissible function has variation at least twice its height, this proves (6.1). It also proves coincidence of maximizing sets: a variation maximizer has height $C_{n,p}$, and every height maximizer has variation between $2C_{n,p}$ and $V_{n,p}$.

The only conclusion still needed for the local theorem is midpoint symmetry. Single-peak geometry by itself does not supply it.

6.3. Smoothness of the point-evaluation profile near two

Let $m = n - 1 \geq 2$, and define

$$K_a(t) = \frac{(a-t)_+^m}{m!}, \quad J(q, a) = \min_{P \in \mathcal{P}_m} \int_0^1 |K_a - P|^q dt. \quad (6.10)$$

By the same moment argument as (6.4), the evaluation norm is $J(q, a)^{1/q}$. We need uniform continuity on the whole interval in a , and twice continuous differentiability in a near $(q, a) = (2, 1/2)$.

First, minimizing polynomials have coefficients bounded uniformly for q in a compact subset of $(1, \infty)$ and $a \in [0, 1]$, by the argument used above. On this bounded coefficient set the integrands in (6.10) vary continuously with (q, a, P) , uniformly in t . Minimizing over that compact set proves continuity of J , including uniform convergence $J(q, \cdot) \rightarrow J(2, \cdot)$ as $q \rightarrow 2$. Strict convexity gives a unique minimizing polynomial; compactness and uniqueness prove continuous dependence of its coefficients.

The following elementary sublevel estimate handles exponents just below two. If P is a polynomial of degree at most m on an interval I of length bounded above and below, and $\|P\|_{L^\infty(I)} \geq b > 0$, then

$$|\{t \in I : |P(t)| \leq h\}| \leq C(m, b, I) h^{1/m} \quad (0 < h < 1). \quad (6.11)$$

The constant is uniform for intervals in the stated length range. To prove this, let the sublevel set have length $\ell > 0$. Divide its mass into sufficiently many equal portions and choose $m+1$ points from alternate portions. They can be chosen in the sublevel set with consecutive separations bounded below by a fixed multiple of $\ell/(m+1)$: coordinate separation is at least the measure of the intervening portion. Lagrange interpolation at these points gives $\|P\|_\infty \leq C_m h \ell^{-m}$, with the interval-length factor absorbed into the constant. Rearrangement gives (6.11); approximation of the chosen points handles essential rather than attained sublevel membership.

Consequently, for $0 < s < 1/m$, integration of distribution functions gives

$$\int_{\{|P| < h\}} |P|^{-s} dt \leq C h^{1/m-s}. \quad (6.12)$$

For example, split the region into dyadic bands $2^{-k-1}h \leq |P| < 2^{-k}h$, apply (6.11) to each, and sum the convergent geometric series. The estimates also show uniform integrability of these inverse powers.

At $(q, a) = (2, 1/2)$, the minimizing residual in (6.10) is a nonzero polynomial on each of the two half-intervals. If it were identically zero on one half, its orthogonality to constants would fail on the other: the remaining residual would be a nonzero constant multiple of the nonnegative truncated power. Continuity of the coefficients therefore supplies a uniform positive lower bound for the sup norm of each polynomial piece on its own interval when (q, a) and the polynomial coefficients are sufficiently close to this point. The intervals' lengths also stay bounded away from zero. Thus (6.11)–(6.12) apply uniformly to both pieces.

Write $P_c(t) = \sum_{j=0}^m c_j t^j$ and $R = K_a - P_c$. The normal equations are

$$\mathcal{N}_j(q, a, c) = \int_0^1 |R|^{q-2} R t^j dt = 0, \quad 0 \leq j \leq m. \quad (6.13)$$

Take a small open neighborhood with $q > 2 - 1/(2m)$. The equations are continuously differentiable in (q, a, c) there. Their derivatives in (a, c) are obtained by multiplying the variation of R by $(q -$

$1)|R|^{q-2}$, and the derivative in q has integrand $|R|^{q-2}R \log |R| t^j$, with value zero at $R = 0$. These assertions follow first away from a small sublevel set of R , by ordinary differentiation. The uniform bound (6.12) makes the omitted derivatives uniformly integrable and arbitrarily small as the sublevel threshold tends to zero. The logarithmic derivative is bounded by a constant times a small positive power of $|R|$ near zero. This also proves continuity of the derivatives by splitting into the same two regions. The moving interface $t = a$ causes no additional term, since K_a and its first a -derivative are continuous there for $m \geq 2$.

At the base point, the coefficient derivative of (6.13) is the negative definite Gram matrix $-\left[\int_0^1 t^{i+j} dt\right]_{i,j=0}^m$. The finite-dimensional implicit function theorem yields a continuously differentiable coefficient map $c(q, a)$. It is the minimizing polynomial map by strict convexity and continuity already established.

The envelope derivative and its second derivative are

$$\begin{aligned} J_a &= q \int |R|^{q-2} R K_a^{[1]}, \\ J_{aa} &= q(q-1) \int |R|^{q-2} (K_a^{[1]} - P_{c_a}) K_a^{[1]} + q \int |R|^{q-2} R K_a^{[2]}, \end{aligned} \tag{6.14}$$

where $K_a^{[1]} = (a-t)_+^{m-1}/(m-1)!$ and $K_a^{[2]} = (a-t)_+^{m-2}/(m-2)!$. For $m = 2$, the last expression is the indicator of $t < a$, with its value at $t = a$ immaterial. The coefficient term in the first derivative vanishes by (6.13). The second derivative exists and is continuous by (6.12), boundedness of these kernels, and dominated convergence away from the moving interface. In particular no claim that all residual roots are simple is needed.

At $q = 2$, the credited point-evaluation formula (4) gives

$$J(2, a) = \frac{[a(1-a)]^{2n-1}}{(2n-1)((n-1)!)^2}. \tag{6.15}$$

It has a unique maximum at $a = 1/2$, with strictly negative second derivative there. By continuity of (6.14), it follows that $J(q, \cdot)$ is strictly concave on one fixed neighborhood of $1/2$, for all q close enough to two. Uniform convergence on $[0, 1]$ forces every global maximum into that neighborhood. Reflection of the original evaluation problem gives $J(q, a) = J(q, 1-a)$. Strict concavity and this symmetry show that $1/2$ is the unique global maximizing point.

6.4. Completion of the proof

Shrink ε_n so that both the variation-maximizer argument and the evaluation-profile argument apply. We have established the constant identity and coincidence of maximizing sets. Every height maximizer must realize evaluation at its absolute maximum, and Section 6.3 forces that point to be $1/2$. The norm-one positive evaluation extremizer is unique by equality in Hölder in (6.4), or by strict convexity of the unit ball of L^p . Reflection preserves the norm and evaluation at $1/2$, so uniqueness makes this function midpoint symmetric. Section 6.2 already gives its strict derivative signs and its unique peak. This proves Theorem 5. $\square$

6.5. The unresolved global step

Compactness away from the endpoints of the exponent range. The argument in Lemma 6 gives the following more general statement. If $p_j \rightarrow p_0 \in (1, \infty)$ and u_j are norm-one variation maximizers,

a subsequence converges in C^n to a norm-one variation maximizer at p_0 , and $V_{n,p_j} \rightarrow V_{n,p_0}$. To see this, keep p_j in any fixed compact subinterval of $(1, \infty)$; the exponents $q_j - 1$ then have a positive lower bound, so (6.6) still gives a common Hölder modulus. The limit has norm one at p_0 . For the lower bound on the limiting optimal value, test a maximizer at p_0 ; (6.6) makes its highest derivative bounded, so it is admissible at all nearby exponents and its varying norms converge. The upper bound follows by applying the p_0 inequality to the limit. This proves the statement without assuming uniqueness or symmetry at p_0 .

Thus a local continuation criterion at another exponent is available: if all its normalized variation maximizers are scalar-sign copies of one single-peak function with nonzero endpoint highest derivatives and a nondegenerate peak, and its point-evaluation profile has a unique midpoint maximum with a continuously persisting negative second derivative, the same argument proves the conjecture in a neighborhood of that exponent. These hypotheses must be established there; they cannot be imported from the case $p = 2$.

The strongest compactness assertion here is C^n convergence of *variation maximizers*, established from their dual optimality equation. It is not asserted for arbitrary normalized Sobolev sequences. This distinction is necessary to preserve the derivative signs near the clamped endpoints.

The point-profile calculation uses finite-dimensional best approximation only after proving uniform control of possible multiple zeros in its polynomial residual. Omitting (6.11)–(6.12) would leave the two-sided neighborhood in p unjustified, since $q < 2$ creates negative powers at residual zeros.

This argument does not prove that the set of valid exponents is closed. At a hypothetical first exponent where the result ceases to hold, compactness of the old single-peak branch does not exclude a different multiple-peak global maximizer with the same value. Theorem 10 below resolves the separate point-evaluation symmetry question at every finite exponent. The possible coexistence of multiple-peak variation maximizers remains a substantive global obstacle.

The remaining task is to exclude such competing branches or to find a counterexample. Theorem 5 supplies a local result, not the full conjecture. No numerical experiment is used in its proof.

7. Complete symmetry at the fourth-order measure endpoint

We first settle the endpoint symmetry problem explicitly in order four by a finite alternation argument. The theorem below concerns the measure class (21); its constants also apply to the ordinary clamped Sobolev space by Theorem 4.

Theorem 8 (fourth-order endpoint). One has

$$C_{4,1} = \frac{1}{1296}, \quad V_{4,1} = \frac{1}{648}. \quad (7.1)$$

The norm-one relaxed extremizers for height and variation are exactly μ_* and $-\mu_*$, where

$$\mu_* = \frac{1}{9}(\delta_0 + \delta_1) - \frac{1}{4}(\delta_{1/6} + \delta_{5/6}) + \frac{5}{18}\delta_{1/2}. \quad (7.2)$$

Their potentials are even about $1/2$. No nonzero extremizer exists in the ordinary $W_0^{4,1}$ class. Thus the full constant, coincidence and symmetry assertions in [NS, Conjecture 4.14] hold for $n = 4$, $p = 1$, with the endpoint understood in the measure sense.

The all-order symmetry assertion is proved separately in Section 8. The point estimates for orders two and three are already included in [NS, Theorem 4.5]; those cases are not new here.

7.1. Extreme measures and a dual cubic

Let C be the maximal relaxed height. It is positive and attained by Section 5. Choose a maximizing point and sign. The corresponding evaluation functional exposes a nonempty compact face of $\mathcal{M}_4$. An extreme point of that face is extreme in $\mathcal{M}_4$ and still maximizes height. By the moment-circuit argument in Theorem 4, after choosing its positive potential it has nodes

$$0 < a < b < c < 1, \quad (t_0, t_1, t_2, t_3, t_4) = (0, a, b, c, 1),$$

$$\mu = \frac{1}{D} \sum_{i=0}^4 \beta_i \delta_{t_i}, \quad \beta_i = \prod_{j \neq i} (t_i - t_j)^{-1}, \quad D = \sum_i |\beta_i|. \quad (7.3)$$

The end nodes are 0 and 1 because stretching a shorter support, with derivative-measure norm fixed, multiplies height by the inverse cube of its length. The signs of the weights in (7.3) are $+, -, +, -, +$.

Write U for this positive potential and let x maximize it. Before a it is a positive multiple of x^3 ; after c it is a positive multiple of $(1-x)^3$. Its derivatives at a and c have strict opposite signs. Hence $a < x < c$. Reflecting if necessary, assume

$$a < x \leq b < c < 1. \quad (7.4)$$

Put $K_x(t) = (x-t)_+^3/6$. There is a polynomial P of degree at most three such that

$$R(t) = K_x(t) - P(t), \quad \|R\|_\infty = C, \quad (R(0), R(a), R(b), R(c), R(1)) = (C, -C, C, -C, C). \quad (7.5)$$

Here are the duality details. The dual of $C[0, 1]/\mathcal{P}_3$ is the space of finite measures annihilating $\mathcal{P}_3$, with total-variation norm. Hahn-Banach and the Riesz representation theorem therefore identify the distance of K_x to $\mathcal{P}_3$ with the norm of its evaluation functional on $\mathcal{M}_4$. That norm equals C , because the chosen measure attains the global height. A best polynomial exists by finite-dimensional coercivity. Equality in $\int R d\mu \leq \|R\|_\infty \|\mu\|_{\text{TV}}$ forces the five values in (7.5). Since R is continuously differentiable, its derivative vanishes at a, b, c .

Let $R_+ = -P$, so that $R = R_+$ on $[x, 1]$. The cubic R_+ has a maximum at b and a minimum at c . Consequently, for some $k > 0$,

$$R'_+(t) = k(t-b)(t-c). \quad (7.6)$$

Equality of its values at b and 1 gives, on putting $h = c - b$,

$$1 - b = \frac{3}{2}h, \quad b = 1 - \frac{3}{2}h, \quad c = 1 - \frac{1}{2}h. \quad (7.7)$$

Indeed $\int_b^1 (t-b)(t-c)dt = (1-b)^2((1-b)/3 - h/2) = 0$.

7.2. The remaining algebraic constraints

On the left of x , $R(t) = R_+(t) + (x-t)^3/6$. The equations $R'(a) = 0$ and $R(a) = R(c)$ yield

$$(x-a)^2 = 2k(b-a)(c-a), \quad (x-a)^3 = k(c-a)^2(2(c-a) - 3h). \quad (7.8)$$

Set $v = (c-a)/h > 1$. Dividing the two equations, and using $0 < x-a \leq b-a$, gives

$$\frac{3}{2} < v \leq 2, \quad a = 1 - h \left(v + \frac{1}{2} \right), \quad x = 1 - \frac{h(2v-1)}{2(v-1)}. \quad (7.9)$$

For clarity, $(x-a)/h = v(2v-3)/(2(v-1))$. Its positivity gives $v > 3/2$, and its upper bound $v-1$ gives $v \leq 2$.

The potential itself is stationary at x . Only the nodes $0, a$ contribute to its first derivative under (7.4), so $U'(x) = 0$ reads

$$(x-a)^2 = x^2 \frac{(b-a)(c-a)(1-a)}{bc}. \quad (7.10)$$

Combining this with (7.8) gives $k = x^2(1-a)/(2bc)$. Also $R(0) = R(1)$ implies

$$\frac{x^3}{6} = k \int_0^1 (t-b)(t-c)dt = \frac{kb^2}{3}, \quad x = \frac{b(1-a)}{c}. \quad (7.11)$$

The integral equals $b^2/3$ by (7.7). All divided quantities are positive.

Substitute (7.7) and (7.9) into (7.10)-(7.11). The resulting two quadratic equations in h are

$$Ah^2 + Bh + C_0 = 0, \quad D_0h^2 + Eh + F = 0, \quad (7.12)$$

where

$$\begin{aligned} A &= 6v^2 - v - 4, & B &= -4v^2 - 4v + 6, \\ C_0 &= 4v - 4, & D_0 &= 2(v-1)(2v-1)^2, \\ E &= -16v^3 + 42v^2 - 31v + 4, & F &= 2v(2v-3)^2. \end{aligned} \quad (7.13)$$

One can check the second equation without elimination software: after substitution, the difference between the two sides of (7.10), multiplied by bc , is

$$(x-a)^2bc - x^2(b-a)(c-a)(1-a) = \frac{ah^2v}{8(v-1)^2}(D_0h^2 + Eh + F). \quad (7.14)$$

The first equation is $4(v-1)(xc - b(1-a)) = 0$. These identities justify (7.12) with no loss from a vanishing factor.

Two quadratics in (7.12) with a common root must satisfy

$$0 = (AF - C_0D_0)^2 - (AE - BD_0)(BF - C_0E) = 16(v-2)^2(v-1)(2v+1)\Pi(v), \quad (7.15)$$

where direct expansion gives

$$\Pi(v) = 8v^6 - 24v^5 - 6v^4 + 64v^3 - 48v^2 + 9v - 2. \quad (7.16)$$

To verify the first equality directly, put $L = AE - BD_0$, $M = AF - C_0D_0$ and $N = BF - C_0E$. Subtracting multiples of the two equations gives $Lh + M = 0$ and $h(Mh + N) = 0$. Since $h > 0$, these imply $M^2 - LN = 0$. The factorization in the second equality is an ordinary polynomial identity.

There is a short sign check for its last factor. For $z = v - 3/2 \in [0, 1/2]$,

$$\begin{aligned}\Pi(3/2 + z) &= 8z^6 + 48z^5 + 84z^4 + 28z^3 - \frac{87}{2}z^2 - 27z - 2 \\ &\leq -2z^2 - 27z - 2 < 0.\end{aligned}\tag{7.17}$$

Indeed the four positive terms are at most $(1/2 + 6 + 21 + 14)z^2 = (83/2)z^2$. Since $v > 3/2$, equation (7.15) forces $v = 2$. Both equations in (7.12) then reduce to $2(3h - 2)(3h - 1) = 0$. The root $h = 2/3$ makes $a < 0$ and is inadmissible. The other gives

$$(a, b, c, x) = \left(\frac{1}{6}, \frac{1}{2}, \frac{5}{6}, \frac{1}{2}\right).\tag{7.18}$$

Thus every globally maximizing extreme measure has precisely these nodes. This conclusion concerns maximizers directly; it does not require a knot-reflection inequality for every configuration.

7.3. Constants and all equality cases

The symmetric dual cubic, after normalization, is the centered cubic Chebyshev spline already described in [dB03, p. 74]. The new issue in this section is unrestricted global optimality and its equality classification. The weights in (7.3), normalized to total variation one at (7.18), are exactly (7.2). Their potential is

$$U_*(x) = \frac{1}{6} \left[\frac{x_+^3 + (x-1)_+^3}{9} - \frac{(x-1/6)_+^3 + (x-5/6)_+^3}{4} + \frac{5}{18}(x-1/2)_+^3 \right].\tag{7.19}$$

It is symmetric and unimodal by Section 5. Substitution at $x = 1/2$ gives $U_*(1/2) = 1/1296$. The extreme maximizing measure exists, and (7.18) identifies it, so this value is the unrestricted sharp height, not merely the maximum within symmetric configurations. Theorem 4 then gives the variation constant $1/648$.

For an arbitrary norm-one height maximizer, expose a face by evaluation at one of its maximum-height points, with the appropriate sign. Every extreme point of that face is a height-maximizing extreme measure and therefore equals μ_* or $-\mu_*$. Only one of these two signs can belong to the chosen face, because the exposed value is positive. Krein-Milman makes the face a singleton. This proves the stated equality classification, and Theorem 4 transfers it to variation. An ordinary Sobolev extremizer would have an absolutely continuous fourth derivative, whereas (7.2) is atomic. Hence ordinary attainment is impossible. The approximation argument in Theorem 4 still recovers both sharp constants. $\square$

8. Symmetry in every order at the measure endpoint

The fourth-order calculation can be replaced by an all-order comparison. The key is to follow the contacts of the best uniform polynomial approximation as the evaluation point moves. Their velocities are strictly less than one. The monotonicity of B-spline peaks then compares the evaluation point with the peak of its own optimizing potential.

Theorem 9 (all-order measure endpoint). For every $n \geq 2$, the point-evaluation norm

$$A_n(x) = \sup_{\mu \in \mathcal{M}_n} U_\mu(x)$$

is strictly increasing on $(0, 1/2)$ and strictly decreasing on $(1/2, 1)$. There is a unique measure μ_n of total variation one attaining positive evaluation at $1/2$. Its potential is positive in $(0, 1)$, symmetric about $1/2$, and has exactly one peak. The normalized relaxed height and variation extremizers are precisely μ_n and $-\mu_n$. In particular, all assertions of [NS, Conjecture 4.14] hold at $p = 1$ for every order, with the endpoint interpreted in the measure sense. The sharp constants in the ordinary clamped Sobolev class agree with the relaxed constants, but no nonzero ordinary extremizer exists.

The full conjecture at the other exponents is not asserted here. The B-spline peak monotonicity used below is credited to Foucart [F]; the endpoint dependence needed here is supplied explicitly.

8.1. A zero count and regular contacts

We first record an elementary zero-count fact. Suppose f is polynomial of degree at most d on each side of a single knot x and belongs to C^{d-1} , with neither polynomial piece identically zero. Then the number of isolated zeros, counted with polynomial multiplicity away from x and the order of the first nonzero common derivative at x , is at most $d + 1$. The only case where no common derivative through order $d - 1$ is nonzero at a knot zero is

$$f(t) = a_-(t - x)^d \quad (t < x), \quad f(t) = a_+(t - x)^d \quad (t > x),$$

with $a_- a_+ \neq 0$; this case has just one distinct zero.

Here is a proof of the count, including missing coefficients. Write the common Taylor coefficients through degree $d - 1$ as $a_0, \dots, a_{d-1}$ and the two top coefficients as a_d^-, a_d^+ . If the first nonzero common coefficient has index r , factor $(t - x)^r$. Descartes' rule of signs bounds the zeros on the right and left by the sign variations in the remaining coefficient list and the list with signs alternating according to degree. For successive nonzero common coefficients whose degrees differ by h , the sum of the two variation contributions is at most h : it is one for odd h , and zero or two for even h . Their total is at most $d - r - 1$. The two independently chosen top coefficients add at most two. Adding the knot multiplicity r gives $d + 1$. If all common coefficients vanish, the displayed monomial case applies. Endpoint zeros of the interval under consideration are among the roots counted by this argument.

Fix $n \geq 3$, put $m = n - 1$, and write

$$K_x(t) = \frac{(x - t)_+^m}{m!}, \quad R_x(t) = K_x(t) - P_x(t), \quad A = A_n(x) = \|R_x\|_\infty, \quad (8.1)$$

where P_x is a best uniform polynomial of degree at most m . The moment duality in Section 7 applies in every order. A maximizing evaluation face has an extreme measure supported on $n + 1$ distinct nodes, and the signed weights force $n + 1$ alternating contacts of R_x .

Neither piece of R_x can be constant: smooth matching at x would make the other a constant plus a multiple of $(t - x)^m$, yielding at most two alternating contacts. Thus the contact set is finite. Every interior contact is a strict local extremum. Applying the zero count to R'_x with $d = n - 2$ shows that it has at most $n - 1$ zeros counting multiplicity. Since there are already at least $n - 1$ interior contacts, the contact set is exactly

$$0 = t_0(x) < t_1(x) < \dots < t_{n-1}(x) < t_n(x) = 1, \quad R_x(t_i) = (-1)^i A. \quad (8.2)$$

All interior critical points are simple, and the endpoint derivatives are nonzero. For $n = 3$, a contact cannot equal x : the two linear pieces of R'_x would otherwise have their common zero at x , and a second critical point would force one piece to vanish identically. For $n \geq 4$, a multiple critical point at x also violates the same zero count. Consequently, in every case,

$$\operatorname{sgn} R''_x(t_i) = (-1)^{i+1} \quad (1 \leq i < n), \quad R'_x(0) < 0, \quad \operatorname{sgn} R'_x(1) = (-1)^n. \quad (8.3)$$

Every measure maximizing positive evaluation at x is supported on this contact set, by equality in the dual norm inequality. Vandermonde rank gives a one-dimensional nullspace for the n moment conditions on $n + 1$ nodes. Positive evaluation and total variation one fix its sign and magnitude. Denote this unique measure by μ_x . Its potential is the positive normalized B-spline on these nodes.

The contacts, polynomial coefficients and A_n are C^1 functions of x . Indeed, differentiate the $n + 1$ contact-value equations and $n - 1$ critical-point equations. The contact-value block in the polynomial coefficients and A is invertible: a nonzero polynomial of degree at most $n - 1$ cannot take nonzero alternating values at $n + 1$ ordered points. The remaining diagonal block consists of the nonzero numbers $R''_x(t_i)$. The implicit function theorem applies. Its equations are C^1 even when x equals a contact for $n \geq 4$; the case $n = 3$ was excluded above. To identify this local branch with the actual optimizer, use uniqueness and compactness: best polynomial coefficients are locally bounded and their subsequential limits are best approximants; the norming measures converge to the unique norming measure, whose $n + 1$ nonzero atoms prevent contact coalescence.

8.2. Every contact moves more slowly than evaluation

A dot denotes differentiation with respect to x ; a prime denotes differentiation with respect to t . Put $\ell = A'_n(x)/A_n(x)$ and define

$$V(t) = \dot{R}_x(t) + R'_x(t) - \ell R_x(t). \quad (8.4)$$

The identity $\partial_x K_x + \partial_t K_x = 0$ shows that V is again a degree-at-most- m spline with one simple knot: it is a polynomial minus ℓK_x . Differentiating (8.2) and using $R'_x(t_i) = 0$ at interior contacts gives

$$V(t_i) = 0 \quad (1 \leq i < n), \quad V(0) = R'_x(0) < 0, \quad V(1) = R'_x(1). \quad (8.5)$$

Neither piece of V vanishes identically, since its endpoint values are nonzero. The zero count gives at most n zeros counting multiplicity. The $n - 1$ prescribed interior zeros must therefore all be simple and there can be no additional zero. To see the parity step explicitly, the endpoint signs in (8.5) require the parity of the number of crossings to be $n - 1$. An additional simple zero or one double prescribed zero would reverse this parity; restoring it costs another zero and exceeds the bound n . A prescribed zero of multiplicity at least three already exceeds that bound. At a knot zero where the first m common coefficients vanish, both pieces would be monomials and there would be only one distinct zero, contrary to the $n - 1 \geq 2$ prescribed zeros. Thus this exception does not interfere with the parity argument.

It follows that $V'(t_i)$ has sign $(-1)^{i+1}$, the same as $R''_x(t_i)$. Differentiating the critical-point equation yields

$$\dot{R}'_x(t_i) + R''_x(t_i)t'_i(x) = 0, \quad V'(t_i) = R''_x(t_i)\{1 - t'_i(x)\}.$$

Hence

$$t'_i(x) < 1 \quad (1 \leq i < n). \quad (8.6)$$

No lower bound on these velocities is needed.

8.3. The dependence of a B-spline peak on all its knots

Let $B_{\mathbf{t}}$ be the conventional positive B-spline of degree $m \geq 2$ on $m+2$ simple knots and let $s(\mathbf{t})$ be its unique peak. Foucart [F, Proposition 6] proves

$$\frac{\partial s}{\partial t_j} > 0 \quad \text{for every interior knot } t_j. \quad (8.7)$$

The peak is nondegenerate and depends differentiably on the knots. We need the same inequality for the two support endpoints. We prove this extension using the same divided-difference identities, keeping its scope separate from the cited statement.

By affine scaling take the support to be $[0, 1]$, and let B_j be the degree- $m+1$ B-spline obtained by repeating the endpoint t_j . The knot-insertion identity in [F, Lemma 3 and Formulae 5] gives

$$B_j(y) = \frac{y - t_j}{m+1} B'_j(y) + B_{\mathbf{t}}(y). \quad (8.8)$$

At $s = s(\mathbf{t})$, differentiating once and eliminating $B'_j(s)$ yields

$$(s - t_j)^2 B''_j(s) = m(m+1) \{B_j(s) - B_{\mathbf{t}}(s)\}. \quad (8.9)$$

For the left endpoint $t_j = 0$, integrating (8.8) with $B_j(1) = 0$ gives

$$B_j(s) = (m+1)s^{m+1} \int_s^1 B_{\mathbf{t}}(y) y^{-m-2} dy \leq B_{\mathbf{t}}(s)(1 - s^{m+1}) < B_{\mathbf{t}}(s). \quad (8.10)$$

For the right endpoint, the reflected formula gives the same strict inequality. Thus $B''_j(s) < 0$. Differentiation of the divided difference defining $B_{\mathbf{t}}$ with respect to an endpoint gives

$$\partial_{t_j} B'_{\mathbf{t}}(s) = -\frac{1}{m+1} B''_j(s) > 0. \quad (8.11)$$

The derivative of the support-length normalization contributes a multiple of $B'_{\mathbf{t}}(s) = 0$, so it contributes nothing to (8.11). Since $B''_{\mathbf{t}}(s) < 0$, implicit differentiation of $B'_{\mathbf{t}}(s) = 0$ proves (8.7) for the endpoints too.

Translation equivariance now gives

$$\sum_{j=0}^{m+1} \frac{\partial s}{\partial t_j} = 1, \quad \frac{\partial s}{\partial t_j} > 0 \quad \text{for every } j. \quad (8.12)$$

8.4. Strict midpoint comparison and equality

Let $s(x) = s(t_0(x), \dots, t_n(x))$ be the peak of U_{μ_x} . The normalizing factor of this potential does not affect its peak. Equations (8.6) and (8.12), together with $t'_0 = t'_n = 0$, show that

$$s'(x) = \sum_{i=1}^{n-1} \frac{\partial s}{\partial t_i} t'_i(x) < \sum_{i=1}^{n-1} \frac{\partial s}{\partial t_i} < 1. \quad (8.13)$$

Reflection and uniqueness of the positive-evaluation norming measure at $1/2$ give $s(1/2) = 1/2$. Therefore $s(x) - x$ is strictly decreasing and

$$s(x) > x \quad (x < 1/2), \quad s(x) < x \quad (x > 1/2). \quad (8.14)$$

Finally, uniqueness and weak-star continuity of μ_x , together with joint continuity of $\partial_x K_x$ for $n \geq 3$, give the envelope identity

$$A'_n(x) = U'_{\mu_x}(x). \quad (8.15)$$

It follows directly by testing the optimizer at each of two neighboring points in the other evaluation problem and taking the two difference quotients. The B-spline potential is strictly increasing before its unique peak and strictly decreasing after it. Equation (8.14) therefore proves the asserted strict monotonicity of A_n .

Every global height maximizer must consequently attain its absolute height at $1/2$. The unique norming measure there, up to the sign of evaluation, is μ_n . Its potential is symmetric by reflection and positive by the moment-circuit formula. Theorem 4 transfers the complete classification to variation. The ordinary constants agree by the recovery argument there; an ordinary extremizer would have an absolutely continuous highest derivative, whereas the unique relaxed one is a nonzero atomic circuit. This rules out ordinary attainment.

For $n = 2$, the direct formula $A_2(x) = x(1 - x)/2$ and its unique three-point norming measure give the same conclusions. This completes the proof. $\square$

9. Point symmetry at every finite exponent

The preceding cancellation also has a finite-exponent form. Instead of following finitely many contacts, we integrate an auxiliary spline against a positive weight and count its zeros. Throughout this section $q = p/(p - 1)$ is the dual exponent; it is not the target exponent in [NS, Conjecture 4.3].

Theorem 10 (finite-exponent point symmetry). Let $n \geq 2$ and $1 < p < \infty$. The point-evaluation norm

$$A_{n,p}(a) = \sup\{u(a) : u \in W_0^{n,p}(0, 1), \|u^{(n)}\|_p \leq 1\} \quad (9.1)$$

is strictly increasing on $(0, 1/2)$ and strictly decreasing on $(1/2, 1)$. The nonzero extremizers for the height inequality in (2) are precisely the scalar multiples of the unique positive norm-one midpoint normer. That normer is symmetric, positive in the interior, and has a unique nondegenerate maximum.

The order-two statement and order-three midpoint symmetry are prior results [W, Theorem 1.1 and Lemma 1.3], recorded also in [NS, Theorem 4.5]. Watanabe et al. explicitly note in [W, Remark 1.4] that their symmetrization proof does not extend to orders at least four. The theorem above gives

the all-order comparison. Together with Theorem 9 and the credited $p = \infty$ theorem [NS, Theorem 4.12], it establishes the $k = 0$, target- L^∞ case of [NS, Conjecture 4.3] for every $n \geq 2$ and $1 \leq p \leq \infty$. It does not establish the remaining variation identity in Conjecture 4.14.

9.1. Simple residual zeros and a single peak

We prove the theorem for $n \geq 3$ and write $m = n - 1$. As in Section 6, let P_a be the unique best L^q polynomial approximant of degree at most m to $K_a(t) = (a - t)_+^m / m!$. Set

$$R_a = K_a - P_a, \quad f_a = |R_a|^{q-2} R_a, \quad u_a(t) = \int_0^t \frac{(t-z)^{n-1}}{(n-1)!} f_a(z) dz. \quad (9.2)$$

The normal equations (6.5) make u_a clamped at both endpoints and give

$$u_a(a) = \int_0^1 |R_a|^q > 0, \quad v_a = \frac{u_a}{\|R_a\|_q^{q-1}}. \quad (9.3)$$

Here v_a is the unique positive-evaluation normer. This follows from the dual formula and equality in Hölder's inequality, as in Section 6; the best-approximation formula itself is credited to [GS, Theorem 2 and Corollary 1].

The residual has at least n sign changes. Otherwise the polynomial having its sign-change points as roots, with a suitable sign, has degree at most $n - 1$ and gives a strictly positive integral against f_a , contradicting (6.5). Neither polynomial piece of R_a vanishes identically: smooth matching would make the other a nonzero multiple of $(t - a)^m$, of constant sign on its side, contradicting the constant moment. The one-knot zero count of Section 8.1 bounds the total number of zeros, with multiplicity, by n . Its exceptional two-monomial case has only one distinct zero and is excluded. Thus R_a has exactly n simple interior zeros, and its endpoint values are nonzero.

Repeated Rolle counting shows that u_a has no interior zero and u'_a has exactly one simple interior zero. Indeed, $u_a \in C^n$ has endpoint zeros of multiplicity n . An additional interior zero would force at least $n + 1$ distinct zeros of f_a after n differentiations. Applied to u'_a , whose endpoint multiplicities are $n - 1$, the same count excludes either two distinct interior zeros or a degenerate zero. Consequently $u_a > 0$ in $(0, 1)$ and its unique peak $s(a)$ satisfies

$$u''_a(s(a)) < 0, \quad f_a(0) > 0, \quad (-1)^n f_a(1) > 0. \quad (9.4)$$

The last two signs follow from the endpoint Taylor expansions.

9.2. Differentiation through the residual zeros

The preceding simple-root result removes the restriction to exponents near two in the regularity argument. Fix $q > 1$ and $a \in (0, 1)$. The residual is C^1 in t , since $m \geq 2$. Its roots are transverse even if one equals the kernel knot. In a small neighborhood of the minimizing polynomial coefficients and evaluation point, there remain exactly n simple roots with inverse slopes uniformly bounded. For $1 < q < 2$, this gives the locally uniform estimate

$$\int_{\{|R| < h\}} |R|^{q-2} dt \leq Ch^{q-1}. \quad (9.5)$$

It follows by changing variables from t to $R(t)$ in disjoint monotone neighborhoods of the roots; outside them $|R|$ is bounded below. For $q \geq 2$ the weight has no singularity. Differentiating the normal equations in a and the polynomial coefficients is therefore justified by the same sublevel splitting as in Section 6.3. The coefficient Jacobian is

$$-(q-1) \left[\int_0^1 |R_a|^{q-2} t^{i+j} dt \right]_{i,j=0}^{n-1}, \quad (9.6)$$

a negative definite Gram matrix. The implicit function theorem and uniqueness give a C^1 minimizing coefficient map for every fixed $q > 1$.

The same estimate shows that $a \mapsto f_a$ is continuously differentiable as a map into L^1 . Its derivative is $(q-1)|R_a|^{q-2}\dot{R}_a$. As a function of t , f_a is absolutely continuous, with derivative $(q-1)|R_a|^{q-2}R'_a$. Integration gives continuous differentiability of $a \mapsto u_a$ into C^{n-1} , as well as $u_a \in W^{n+1,1}$. The nondegenerate peak in (9.4) therefore depends continuously differentiably on a . In what follows, a dot denotes the a derivative and a prime the t derivative.

9.3. A weighted comparison for peak motion

Fix a and abbreviate $u = u_a$, $R = R_a$, $f = f_a$, and $s = s(a)$. Define

$$\ell = \frac{\dot{u}(s)}{u(s)}, \quad W = \dot{u} + u' - \ell u, \quad H = \dot{R} + R'. \quad (9.7)$$

The identity $\partial_a K_a + \partial_t K_a = 0$ makes H a polynomial of degree at most $n-1$. Almost everywhere,

$$D^n W = |R|^{q-2} T, \quad T = (q-1)H - \ell R. \quad (9.8)$$

The function T is a degree-at-most- $n-1$ spline with one simple knot. If both pieces are nonzero, Section 8.1 bounds its isolated zeros by n , with the two-monomial exception having only one distinct zero. The weight in (9.8) is positive almost everywhere and integrable.

The function $W \in W^{n,1} \subset C^{n-1}$ satisfies

$$\begin{aligned} W^{(j)}(0) &= W^{(j)}(1) = 0 & (0 \leq j \leq n-2), \\ W^{(n-1)}(0) &= f(0) > 0, & W^{(n-1)}(1) = f(1), \\ W(s) &= 0. \end{aligned} \quad (9.9)$$

In particular, W is positive near zero and negative near one. We claim

$$W'(s) < 0. \quad (9.10)$$

Here is a zero-count proof that includes the possible singularities of the weight. If $W'(s) > 0$, the opposite endpoint signs force another root on each side of s . If $W'(s) = 0$ and $W''(s) \neq 0$, there is a double root and another root. If $W'(s) = W''(s) = 0$, there is a root of multiplicity at least three. These derivatives exist because $n \geq 3$. Thus failure of (9.10) supplies interior zeros of total multiplicity at least three, besides the two endpoint zeros of multiplicity $n-1$.

Apply ordinary Rolle's theorem $n-1$ times. It yields at least $n+2$ distinct zeros of $W^{(n-1)}$: the initial count is at least $2n+1$, and all retained multiplicities are exhausted by this stage. On each of the $n+1$ intervening disjoint intervals, the integral of $|R|^{q-2}T$ is zero. If both pieces of T are nonzero,

it is not identically zero on any interval. Positivity of the weight then forces a sign change of T on each interval, contradicting its zero bound.

If one piece of T vanishes identically, smooth matching makes the other a constant multiple of $(t-a)^{n-1}$. Then $W^{(n-1)}$ is the nonzero constant $f(0)$ or $f(1)$ on the zero side and is strictly monotone on the other side. It has at most one zero, again a contradiction. If both pieces vanish, it is constant nonzero and has none. This proves (9.10).

Differentiate $u'_a(s(a)) = 0$. Equations (9.7) and (9.10) give

$$W'(s) = u''_a(s)\{1 - s'(a)\} < 0. \quad (9.11)$$

Since $u''_a(s) < 0$, we obtain $s'(a) < 1$ for every $a \in (0, 1)$.

9.4. Strict midpoint maximization and the remaining gap

Reflection and uniqueness give $s(1/2) = 1/2$. Thus $s(a) - a$ is strictly decreasing, positive before the midpoint and negative after it. The envelope identity is

$$A'_{n,p}(a) = \frac{u'_a(a)}{\|R_a\|_q^{q-1}}. \quad (9.12)$$

The strict single-peak signs of u_a prove the stated monotonicity. Any height extremizer must attain its absolute maximum at $1/2$; uniqueness of the midpoint normer gives all equality cases, and reflection gives its symmetry. This proves Theorem 10. $\square$

Two useful boundary identities follow from the same cancellation. With $J(a) = \int_0^1 |R_a|^q$, integration by parts and the normal equations give

$$\begin{aligned} J'(a) &= |R_a(0)|^q - |R_a(1)|^q, \\ ((n-1)q+1)J(a) &= a|R_a(0)|^q + (1-a)|R_a(1)|^q. \end{aligned} \quad (9.13)$$

For the second identity use $(a-t)R'_a = -mR_a$ plus a polynomial of degree at most m . The theorem therefore also proves the strict ordering of the endpoint residual magnitudes on either side of the midpoint.

This argument classifies height extremizers for every finite exponent. It does not imply that every variation maximizer has a single peak. Its dual residual in (6.6) comes from H_g , which may have many changes of polynomial piece when $g = \text{sign}(u')$. The one-knot bound in (9.8) is then unavailable. This is the remaining obstruction to the full variation conjecture.

10. Variation continuation from the measure endpoint

The unique endpoint measure also controls arbitrary variation maximizers as the input exponent decreases to one. Two issues require care: ordinary highest derivatives develop concentration, and in order three weak measure convergence does not directly give uniform second-derivative convergence.

Theorem 11 (continuation above one). For every $n \geq 3$ there exists $\eta_n > 0$ such that, for $1 < p < 1 + \eta_n$,

$$V_{n,p} = 2C_{n,p}. \quad (10.1)$$

The nonzero extremizers in the two inequalities are exactly the scalar multiples of the unique positive norm-one midpoint normer. They are symmetric and strictly unimodal up to sign. The interval width is not estimated, and overlap with the interval in Theorem 5 is not asserted. The case $n = 2$ is already known for all p by [NS, Remark 4.15].

10.1. Compactness of arbitrary endpoint sequences

Fix $n \geq 3$. Let $p_j > 1$ tend to one and let u_j be any norm-one variation maximizers, whose existence was proved in Section 6. Write $f_j = u_j^{(n)}$ and $\mu_j = f_j dt$. Since the interval has length one,

$$\|\mu_j\|_{\text{TV}} = \|f_j\|_1 \leq \|f_j\|_{p_j} = 1.$$

A subsequence converges weak-star to some $\mu \in \mathcal{M}_n$. For $n \geq 3$ the potential kernel and its first derivative are jointly continuous on the square. Their integrals over the unit measure ball are uniformly equicontinuous. Pointwise weak-star convergence and this equicontinuity therefore give $u_j \rightarrow U_\mu$ in $C^1[0, 1]$.

The constants satisfy $V_{n,p_j} \leq V_{n,1}$. Conversely, given $\delta > 0$, the recovery argument in Theorem 4 supplies one nonzero smooth clamped function v with

$$\frac{\|v'\|_1}{\|v^{(n)}\|_1} > V_{n,1} - \delta.$$

Its highest-derivative L^{p_j} norm tends to its L^1 norm. Testing this fixed function and then letting δ decrease to zero proves $V_{n,p_j} \rightarrow V_{n,1}$. Thus U_μ maximizes endpoint variation. Theorems 4 and 9 identify μ with μ_n or $-\mu_n$. After choosing the common sign along the subsequence, we have

$$\mu_j \xrightarrow{*} \mu_n, \quad u_j \rightarrow U_{\mu_n} \quad \text{in } C^1, \quad \|\mu_j\|_{\text{TV}} \rightarrow 1. \quad (10.2)$$

The last limit follows from lower semicontinuity and the upper bound one.

We also need convergence of the absolute measures:

$$|\mu_j| \xrightarrow{*} |\mu_n|. \quad (10.3)$$

Indeed, any weak-star limit ν of a subsequence of $|\mu_j|$ dominates $|\mu_n|$. To see this, for nonnegative continuous h and continuous ϕ with $|\phi| \leq h$, pass to the limit in $|\int \phi d\mu_j| \leq \int h d|\mu_j|$ and use the dual description of variation. Both ν and $|\mu_n|$ have mass one by (10.2), so they agree. Every subsequence has this same limit, proving (10.3).

10.2. The limiting dual residual fixes the endpoint signs

Put $U = U_{\mu_n}$ and let $g_j = \text{sign}(u'_j)$, with any choice in $[-1, 1]$ where $u'_j = 0$. The strict signs of U' imply pointwise convergence except at $0, 1/2, 1$ to the midpoint sign function g_* , which is positive before

1/2 and negative after it. Dominated convergence gives $g_j \rightarrow g_*$ in L^1 . Hence $H_j = H_{g_j}$ converges uniformly to

$$H_*(t) = 2K_{1/2}(t) - \frac{(1-t)^{n-1}}{(n-1)!}. \quad (10.4)$$

Let P_j minimize $\|H_j - P\|_{q_j}$ over polynomials of degree at most $n-1$, where $q_j = p_j/(p_j-1) \rightarrow \infty$. The bound $\|P_j\|_1 \leq \|P_j\|_{q_j} \leq 2\|H_j\|_\infty$ and finite-dimensional norm equivalence give bounded polynomial coefficients. Take any subsequential coefficient limit P_∞ . For each fixed finite r , each polynomial P in the same space, and all sufficiently large j ,

$$\|H_j - P_j\|_r \leq \|H_j - P_j\|_{q_j} \leq \|H_j - P\|_{q_j} \leq \|H_j - P\|_\infty. \quad (10.5)$$

First let j tend to infinity, using uniform convergence, and then let r tend to infinity. Thus P_∞ is a best uniform approximant to H_* . Uniqueness of best uniform polynomial approximation and (10.4) identify

$$R_j := H_j - P_j \longrightarrow 2R_{1/2} \quad \text{uniformly,} \quad f_j = \text{sign}(R_j) \left| \frac{R_j}{V_{n,p_j}} \right|^{q_j-1}. \quad (10.6)$$

Here $R_{1/2}$ is the truncated-power residual in Section 8; every subsequence has the same limit. The second identity is the dual optimality equation (6.6).

The endpoint residual values in (8.2) are nonzero, positive at zero and of sign $(-1)^n$ at one. Uniform convergence therefore fixes these signs for R_j on small endpoint neighborhoods, independently of j once it is large. Equation (10.6) fixes the same strict signs for f_j , although its magnitude need not remain bounded. Integration from the clamped endpoints gives $u'_j > 0$ near zero and $u'_j < 0$ near one. On compact sets separated from the endpoints and midpoint, uniform convergence of u'_j to U' preserves the required signs as well.

10.3. Curvature at the midpoint and completion

For $n \geq 4$, the twice-differentiated potential kernel is jointly continuous. The compactness argument above then gives $u''_j \rightarrow U''$ uniformly. The B-spline peak is nondegenerate, as used in Section 8 and [F, Proposition 6]. Thus $u''_j < 0$ on a fixed central interval for all sufficiently large j .

For $n = 3$, the unique symmetric measure has four distinct knots $0, a, 1-a, 1$, with $a < 1/2$. The midpoint lies in a knot-free interval, where U'' is a strictly negative constant. Choose a closed central subinterval $I = [c, d]$ entirely inside this gap. Equation (10.3) and $|\mu_n|(I) = 0$ imply $|\mu_j|(I) \rightarrow 0$. Also $\mu_j([0, c]) \rightarrow \mu_n([0, c])$: continuous approximation of the interval indicator, together with (10.3), justifies this passage because its relative boundary $\{c\}$ has zero limiting variation mass. The endpoint atom at zero is included on both sides. Since $u''_j(x) = \mu_j([0, x])$, we obtain

$$\begin{aligned} \sup_{x \in I} |u''_j(x) - U''(x)| &\leq |\mu_j([0, c]) - \mu_n([0, c])| + |\mu_j|(I) \\ &\longrightarrow 0. \end{aligned} \quad (10.7)$$

Hence the middle curvature is strictly negative in order three too.

Combine this curvature with the derivative signs already proved near the endpoints and on the intervening compact intervals. Each u_j has exactly one critical point for large j , with positive derivative

before it and negative derivative after it. Because the original sequence consisted of arbitrary variation maximizers, a contradiction-sequence argument supplies $\eta_n > 0$ for which every normalized maximizer has this geometry, up to sign, whenever $1 < p < 1 + \eta_n$.

Such a maximizer has variation twice its height, so $V_{n,p} \leq 2C_{n,p}$. The reverse bound always holds. A variation maximizer is therefore a height maximizer, and a height maximizer is a variation maximizer. Theorem 10 gives the midpoint symmetry and unique equality class. This proves Theorem 11. $\square$

11. Finite switches and the stability matrix

The remaining variation problem can be reduced to finite, regular switching configurations without assuming this regularity in advance.

Theorem 12 (regularity of variation maximizers). Fix $n \geq 3$ and $1 < p < \infty$, and put $q = p/(p-1)$. Every norm-one variation maximizer u has finitely many interior critical points, all nondegenerate. If their number is r , then $r \geq 1$ and its dual residual $R = H_{\text{sign}(u')} - P$ has exactly $n + r - 1$ simple interior zeros. Moreover,

$$|u^{(n)}(0)|^p = |u^{(n)}(1)|^p = (n-1)q + 1. \quad (11.1)$$

The number r is uniformly bounded when p ranges over a compact subinterval of $(1, \infty)$, with n fixed. No explicit bound or assertion $r = 1$ is made.

11.1. Endpoint identities without a switching assumption

Write $m = n - 1$, $V = V_{n,p}$, $f = u^{(n)}$ and $g = \text{sign}(u')$. By Section 6,

$$\|R\|_q = V, \quad f = \frac{|R|^{q-2}R}{V^{q-1}}, \quad \int_0^1 Rf = V. \quad (11.2)$$

In particular $R \in W^{m,\infty}$, f is continuous and $u \in C^m$. For some constant c , differentiation of the load gives

$$g - c = (-1)^m R^{(m)} \quad \text{almost everywhere.} \quad (11.3)$$

The chain rule for $|u'|$ is valid regardless of its zero set. It gives

$$\int_0^1 (g - c)u'' = 0, \quad \int_0^1 t(g - c)u'' = -V.$$

Integrate by parts $m - 1$ times. All boundary terms vanish by the clamped conditions. In the second identity use $D^{m-1}(tu'') = tf + (m-1)u^{(m)}$ and $\int R'u^{(m)} = -\int Rf = -V$. The result is

$$\int_0^1 R'f = 0, \quad \int_0^1 tR'f = mV. \quad (11.4)$$

The absolutely continuous function $|R|^q$ has derivative $qV^{q-1}fR'$. Therefore

$$|R(0)|^q = |R(1)|^q = (mq + 1)V^q. \quad (11.5)$$

Equation (11.1) follows from (11.2), since $(q-1)p = q$. In particular f is bounded away from zero on endpoint neighborhoods. No derivative of the sign function g was taken.

11.2. A degenerate critical point is unstable

Suppose $u'(a) = u''(a) = 0$ at an interior point. By (11.1), choose open intervals E_- to its left and E_+ to its right on which $|f|$ has a positive lower bound. There exists $\phi \in C_c^\infty(E_- \cup E_+)$ such that

$$\int_0^1 t^j \phi(t) dt = 0 \quad (0 \leq j < n), \quad \int_0^1 \frac{(a-t)_+^{n-2}}{(n-2)!} \phi(t) dt \neq 0. \quad (11.6)$$

Indeed, if the last functional vanished on the common kernel of the moment functionals, finite-dimensional linear algebra would express it as their linear combination. Testing smooth functions supported in the two intervals would make its kernel a polynomial of degree at most $n-1$ there. That polynomial vanishes on E_+ and hence identically, but the kernel is nonzero on E_- , a contradiction.

Let $h = I^n \phi$. The moment conditions make it a smooth clamped function, and $h'(a) \neq 0$. Since $h^{(n)}$ is supported where $|f|$ is bounded below, the map $\varepsilon \mapsto \|f + \varepsilon \phi\|_p$ is twice continuously differentiable near zero, including when $p < 2$. Applying the sharp variation inequality to $u + \varepsilon h$ and $u - \varepsilon h$ and averaging gives

$$\frac{\|(u + \varepsilon h)'\|_1 + \|(u - \varepsilon h)'\|_1}{2} - \|u'\|_1 \leq C\varepsilon^2. \quad (11.7)$$

The scalar identity $(|y+z| + |y-z|)/2 - |y| = (|z| - |y|)_+$ identifies the left side with $\int (\varepsilon|h'| - |u'|)_+$ for $\varepsilon > 0$. But $u'(t) = o(|t-a|)$ and $|h'|$ is bounded below near a . For every fixed $M > 0$, integration over $|t-a| < M\varepsilon$ gives a lower bound $c_0 M \varepsilon^2$ for all sufficiently small ε , where $c_0 > 0$ is independent of M . Choosing M arbitrarily large contradicts (11.7). Thus $u''(a) \neq 0$ at every interior zero of u' .

All these zeros are isolated. The clamped Taylor expansions and nonzero endpoint values of f fix a nonzero sign of u' near each endpoint. An infinite set of zeros in the remaining compact interval would have an accumulation point, necessarily with $u' = u'' = 0$. Hence there are finitely many zeros. Each is a sign switch, and at least one occurs because $\int_0^1 u' = 0$ and u is nonzero.

11.3. The residual zeros are also simple

Let the critical points be $\tau_1 < \dots < \tau_r$. The moment equations force at least n sign changes of R . Repeated mean-value arguments then force $R^{(m)}$ to take both positive and negative values on sets of positive measure. Equation (11.3), with g taking only the values ± 1 almost everywhere, implies $-1 < c < 1$.

Consequently R has degree exactly m on each of the $r+1$ knot intervals. Its derivative $R^{(m-1)}$ is continuous piecewise affine, with a nonzero slope on each interval, and has at most $r+1$ distinct zeros. The r interior zeros of u' , together with its endpoint zeros of exact multiplicity m , force at least $m+r$ distinct zeros of f after m applications of Rolle's theorem. These are also zeros of R .

Conversely, applying Rolle $m-1$ times to R bounds its distinct zeros by $m+r$. A multiple root would add another zero in this count and force at least $r+2$ distinct zeros of $R^{(m-1)}$, which is impossible. This argument uses only a double root and continuity of R' , including at a switching knot when $m=2$. Thus the residual has exactly $n+r-1$ simple interior zeros.

For the uniform bound, suppose p_j stays in a compact subinterval of $(1, \infty)$ and the corresponding maximizing functions have unbounded critical-point counts. After a subsequence $p_j \rightarrow p_0$, the compactness argument in Section 6.5 gives convergence in C^n to a normalized maximizer at p_0 . Its endpoint highest derivatives are nonzero and all its interior critical points are nondegenerate. Hence the endpoint signs persist, each interior critical point has exactly one nearby critical point for large j ,

and the intervening compact sets contain none. The critical-point counts are eventually equal to that of the limit, a contradiction. This completes the proof of Theorem 12. $\square$

11.4. A justified second-variation condition

Let σ_i be the sign of u' just before τ_i , so the load jumps there by $-2\sigma_i$. Nearby ordered switching points define nearby sign loads. Modulo a polynomial of degree at most $n - 1$, their dual load is

$$H_\tau = 2 \sum_{i=1}^r \sigma_i K_{\tau_i}.$$

Put $J(\tau) = \min_P \int |H_\tau - P|^q$. At the maximizing configuration set

$$w = |R|^{q-2}, \quad L_i(t) = \frac{(\tau_i - t)_+^{n-2}}{(n-2)!}, \quad B_i = L_i - \Pi_w L_i, \quad (11.8)$$

where Π_w is orthogonal projection onto $\mathcal{P}_{n-1}$ in the weighted inner product $\langle v, z \rangle_w = \int w v z$. The weight is positive almost everywhere and integrable. By Theorem 12 all residual roots are simple, so the transverse-root estimates of Section 9 justify continuously differentiating the normal equations in the switching parameters. Their coefficient Gram matrix is positive definite. The same sublevel argument and the envelope formula make J twice continuously differentiable near this configuration.

With the unnormalized potential $U = I^n(|R|^{q-2}R)$, differentiation gives

$$\begin{aligned} J_i &= 2q\sigma_i U'(\tau_i) = 0, \\ J_{ij} &= 4q(q-1)\sigma_i\sigma_j \langle B_i, B_j \rangle_w + 2q\sigma_i\delta_{ij}U''(\tau_i). \end{aligned} \quad (11.9)$$

For $n = 3$, the second derivative of a moving kernel is an indicator. Its contribution is still justified by dominated convergence against the bounded signed residual power; no boundary term is introduced. The weighted term is controlled by the integrable transverse-root estimates.

Define $D_i = -\sigma_i U''(\tau_i) > 0$ and $G_{ij} = \sigma_i\sigma_j \langle B_i, B_j \rangle_w$. Every nearby sign load has dual norm at most V , so the Hessian in (11.9) must be negative semidefinite. Equivalently,

$$2(q-1)G \preceq \text{diag}(D_1, \dots, D_r), \quad (11.10)$$

where $\preceq$ denotes comparison of quadratic forms. This is a necessary condition for every finite-exponent variation maximizer. Theorem 12 justifies its finite, nondegenerate setting; Theorem 13 below excludes $r = 2$. It remains to exclude every configuration with $r \geq 3$ satisfying it, or prove that such a configuration has a smaller value than the midpoint load. No such general exclusion is asserted here.

12. Two switches are unstable

The translation direction in the Hessian has a stronger consequence than the finite Hilbert diagnostic: it excludes a two-switch maximizer at every finite exponent, without a symmetry assumption.

Theorem 13 (two-switch exclusion). Let $n \geq 3$ and $1 < p < \infty$. A nonzero global variation maximizer cannot have exactly two interior critical points. Consequently its critical-point count is either one or at least three.

12.1. A polynomial cancellation under translation

Suppose otherwise, and write the critical points as $a < b$. Choose the sign so that $u' > 0$ near zero. Put $m = n - 1$, $q = p/(p - 1)$, $F = |R|^{q-2}R$ and $U = I^n F$, so U is a positive multiple of u . By Theorem 12, the sign load is $+1, -1, +1$ on the three successive intervals, all residual roots are simple, and

$$F(0) > 0, \quad F(1) = (-1)^{n+1}F(0). \quad (12.1)$$

The signs follow from the $n + 1$ residual roots, and equality of the endpoint magnitudes follows from (11.5).

Translate both switches to $a + s, b + s$. Modulo $\mathcal{P}_m$, the load is $H_s = 2K_{a+s} - 2K_{b+s}$. Let $R_s = H_s - P_s$ be its best L^q residual and $U_s = I^n(|R_s|^{q-2}R_s)$. A dot denotes differentiation in s at zero. Since differentiation in s cancels differentiation in the integration variable for each translated kernel,

$$T = \dot{R} + R' = -\dot{P} - P' \in \mathcal{P}_m. \quad (12.2)$$

Define

$$W = \dot{U} + U', \quad W^{(n)} = (q - 1)|R|^{q-2}T. \quad (12.3)$$

The simple-root estimates used in Section 11 justify differentiation of the signed residual power in L^1 , even when $1 < q < 2$. They also give $F' = (q - 1)|R|^{q-2}R' \in L^1$. Thus $W \in W^{n,1}(0, 1) \subset C^{n-1}[0, 1]$, and (12.3) is an almost-everywhere identity with a positive almost-everywhere integrable weight.

The moment equations make every U_s clamped. Their differentiated equations make $\dot{U}$ clamped as well. Therefore

$$\begin{aligned} W^{(j)}(0) &= W^{(j)}(1) = 0 & (0 \leq j \leq n - 2), \\ W^{(n-1)}(0) &= F(0), & W^{(n-1)}(1) = F(1). \end{aligned} \quad (12.4)$$

12.2. The auxiliary function has a single peak

Put $z = W'$. Its endpoint zeros have exact multiplicity $n - 2$. Equations (12.1) and (12.4) show that $z > 0$ just after zero and $z < 0$ just before one. We claim that z has exactly one interior zero.

Here is the required weighted Rolle count. If selected interior zeros of z have total multiplicity N , counting a zero with $z' = 0$ twice, repeated Rolle arguments through $n - 2$ derivatives give at least $n - 2 + N$ distinct zeros of $W^{(n-1)}$. Between consecutive zeros the integral of $|R|^{q-2}T$ vanishes. Positivity of the weight forces a root of the polynomial T in every such open interval. The polynomial is nonzero: otherwise W would be a polynomial of degree at most $n - 1$ with endpoint zeros of multiplicity $n - 1$ at both ends, contradicting $F(0) > 0$. Hence

$$n - 3 + N \leq \deg T \leq n - 1, \quad N \leq 2. \quad (12.5)$$

This reasoning uses an integral between zeros of the absolutely continuous function $W^{(n-1)}$; it does not differentiate the weight or assign a finite value to it at a residual zero.

There must be a sign-changing zero of z because its endpoint signs are opposite. Two distinct sign-changing zeros would require a third to give these endpoint signs. Any additional tangency contributes multiplicity at least two and, together with the sign-changing zero, also violates (12.5). An infinite zero set is excluded by selecting three distinct zeros. Thus z has exactly one interior zero c , and

$$W' > 0 \quad (0, c), \quad W' < 0 \quad (c, 1). \quad (12.6)$$

In particular W is strictly unimodal and positive in the interior.

12.3. Orthogonality gives a positive second variation

Since $T \in \mathcal{P}_m$, the moment equations give $\int FT = 0$, and (12.3) gives $\int RW^{(n)} = 0$. Integrating by parts $n - 1$ times, the only possible boundary term is $[RW^{(n-1)}]_0^1 = [RF]_0^1 = 0$ by (11.5). Using (11.3) and $W(0) = W(1) = 0$, we obtain

$$0 = \int_0^1 RW^{(n)} = \int_0^1 gW' = 2\{W(a) - W(b)\}. \quad (12.7)$$

Strict unimodality and $a < b$ now imply $a < c < b$. Let $J(s) = \int |R_s|^q$. The envelope identity near zero and its derivative at zero give

$$\begin{aligned} J'(s) &= 2q\{U'_s(a+s) - U'_s(b+s)\}, \\ J''(0) &= 2q\{W'(a) - W'(b)\} > 0. \end{aligned} \quad (12.8)$$

Every nearby translated sign load has dual norm at most the sharp variation norm attained at $s = 0$. The strict positivity in (12.8) contradicts this local maximum, proving Theorem 13. $\square$

For three or more switches, the same translation cancellation gives an alternating sum in (12.7). It no longer equates the values at just two points, so the sign conclusion in (12.8) does not follow. No exclusion of all such configurations is asserted.

13. Affine motion and the three-switch obstruction

The dilation direction complements common translation. It excludes a midpoint-symmetric three-switch maximizer and imposes a further necessary geometric condition on every remaining three-switch candidate.

Theorem 14 (three-switch dilation test). Let $n \geq 3$ and $1 < p < \infty$. A global variation maximizer that is symmetric about the midpoint cannot have exactly three interior critical points. More generally, if a global maximizer has exactly three critical points $a < b < c$, the unique minimum of the auxiliary function for dilation about b , defined below, must lie outside the closed interval $[a, c]$.

13.1. The general affine cancellation

Use the notation of Section 11, put $m = n - 1$, $d = mq + 1$, and set $F = |R|^{q-2}R$, $U = I^n F$. Fix an affine velocity $v(t) = A + Bt$ and move the switches along $\tau_i(s) = \tau_i + sv(\tau_i)$. A dot denotes differentiation at zero. The kernel identity

$$v(\tau)\partial_\tau K_\tau(t) + v(t)\partial_t K_\tau(t) = mBK_\tau(t)$$

shows that

$$T_v = \dot{R} + vR' - mBR \in \mathcal{P}_m. \quad (13.1)$$

Define

$$W_v = \dot{U} + vU' - BdU, \quad W_v^{(n)} = (q-1)|R|^{q-2}T_v. \quad (13.2)$$

As in Section 12, simple residual roots justify parameter and spatial differentiation of the signed residual power in L^1 . Thus $W_v \in W^{n,1}$, its endpoint derivatives through order $n-2$ vanish, and

$$W_v^{(n-1)}(0) = v(0)F(0), \quad W_v^{(n-1)}(1) = v(1)F(1). \quad (13.3)$$

Polynomial orthogonality gives $\int RW_v^{(n)} = 0$. After $n-1$ integrations by parts, the surviving boundary term is $[vRF]_0^1 = B|R(0)|^q$. Hence

$$2 \sum_i \sigma_i W_v(\tau_i) = -B|R(0)|^q. \quad (13.4)$$

Let $\mathcal{H}$ be the Hessian in (11.9), and let $\mathbf{v} = (v(\tau_i))_i$. Since $U'(\tau_i) = 0$, differentiating the envelope formula gives

$$(\mathcal{H}\mathbf{v})_i = 2q\sigma_i W_v'(\tau_i), \quad \mathbf{v}^T \mathcal{H}\mathbf{v} = 2q \sum_i \sigma_i v(\tau_i) W_v'(\tau_i). \quad (13.5)$$

Indeed $W_v' = \dot{U}' + vU'' + B(1-d)U'$, and the last term vanishes at the switches. The switch path is affine in s , so there is no acceleration term.

13.2. Dilation about the middle switch

Suppose there are exactly three switches $a < b < c$, and choose the sign so that $F(0) > 0$. Their signs are $+, -, +$. The $n+2$ simple residual zeros and equal endpoint magnitudes give $F(1) = (-1)^n F(0)$. Take $v(t) = t - b$. Equation (13.3) implies that $W_v' < 0$ near zero and $W_v' > 0$ near one, with endpoint zeros of exact multiplicity $n-2$.

The weighted Rolle argument in (12.5) applies verbatim to (13.2): W_v' has at most two interior zeros counted with the same multiplicity convention. Its opposite endpoint signs exclude a second zero or tangency. Consequently there is a unique interior point z such that

$$W_v' < 0 \quad (0, z), \quad W_v' > 0 \quad (z, 1). \quad (13.6)$$

The auxiliary function is negative in the interior and has its unique minimum at z . If $a \leq z \leq c$, the middle switch has zero velocity and (13.5) becomes

$$\mathbf{v}^T \mathcal{H}\mathbf{v} = 2q\{(a-b)W_v'(a) + (c-b)W_v'(c)\} > 0. \quad (13.7)$$

Both summands are positive when $a < z < c$. At either endpoint of $[a, c]$, one vanishes and the other remains positive. This contradicts maximality. Any remaining three-switch global maximizer therefore requires $z < a$ or $z > c$.

13.3. Midpoint symmetry excludes three switches

If $u(1-t) = u(t)$, its three simple critical points are $a, 1/2, 1-a$, with $a < 1/2$. Dilation about $1/2$ preserves this symmetry of the switch set. Every moved sign load is antisymmetric. Reflection and uniqueness of best L^q polynomial approximation give $R_s(1-t) = (-1)^n R_s(t)$; the clamped potentials U_s are consequently symmetric. Thus $\dot{U}$, $(t-1/2)U'$ and W_v are symmetric as well.

The unique minimum in (13.6) is therefore at $z = 1/2$, between the outer switches. In particular,

$$\mathbf{v}^T \mathcal{H} \mathbf{v} = 4q(1/2 - a)W'_v(1-a) > 0, \quad (13.8)$$

contradicting maximality and completing the proof of Theorem 14. $\square$

The theorem does not establish symmetry of arbitrary variation maximizers. For a nonsymmetric three-switch candidate, it remains to exclude the response-minimum locations $z < a$ and $z > c$, or find another positive Hessian direction. Configurations with at least four switches also remain open.

14. A limitation of weighted Hilbert reduction

The constant derivative weight in Theorem 1 cannot be replaced by an arbitrary positive weight. This matters for a possible reduction of the remaining finite-exponent problem. For a positive bounded weight a bounded away from zero, put

$$C(a) = \sup_{u \neq 0} \frac{\|u\|_\infty}{E_a(u)^{1/2}}, \quad V(a) = \sup_{u \neq 0} \frac{\|u'\|_1}{E_a(u)^{1/2}}, \quad E_a(u) = \int_0^1 a(t)|u'''(t)|^2 dt, \quad (14.1)$$

where the suprema run over $W_0^{3,2}(0,1)$.

Theorem 15 (failure of the arbitrary-weight extension). There exists a positive bounded piecewise-constant weight a , bounded away from zero, such that

$$V(a)^2 > \frac{967}{10^6}, \quad 4C(a)^2 < \frac{900}{10^6}. \quad (14.2)$$

Such weights also exist in $C^\infty([0,1])$. In particular, the arbitrary-weight version of $V = 2C$ and the assertion that every weighted variation maximizer is single-peaked are false. This is a weighted counterexample; it does not disprove [NS, Conjecture 4.14].

14.1. Dual formulas and an explicit weight

Write $\rho = 1/a$ and $\langle f, h \rangle_\rho = \int_0^1 \rho f h$. Let P_ρ be the orthogonal projection onto $\mathcal{P}_2$ in this inner product. The endpoint conditions on u are equivalent to the three moments $\int_0^1 t^j u'''(t) dt = 0$, $0 \leq j \leq 2$. Weighted Hilbert duality therefore gives

$$4C(a)^2 = \max_{0 \leq x \leq 1} D(x), \quad D(x) = \|(I - P_\rho)h_x\|_\rho^2, \quad h_x(t) = (x-t)_+^2. \quad (14.3)$$

For a sign load with switches $b < c$ and successive signs $+, -, +$, integration of the load against the derivative kernel gives $h_b - h_c$ modulo $\mathcal{P}_2$. Hence

$$V(a)^2 \geq J, \quad J = \|(I - P_\rho)(h_b - h_c)\|_\rho^2. \quad (14.4)$$

These formulas have no numerical approximation. For example, if $R = (I - P_\rho)(h_b - h_c)$ and $u''' = \rho R$ with zero initial derivatives, the moment equations give the right endpoint conditions, $E_a(u) = J$, and $\int g u' = J$. Thus $\|u'\|_1 / E_a(u)^{1/2} \geq \sqrt{J}$.

Take $b = 3/7$, $c = 6/7$ and

$$\rho(t) = \frac{1}{100} + \sum_{j=1}^6 \frac{m_j}{2\varepsilon} \mathbf{1}_{[c_j - \varepsilon, c_j + \varepsilon]}(t), \quad \varepsilon = \frac{1}{1000}, \quad (14.5)$$

with

$$(c_1, \dots, c_6) = \frac{1}{100}(3, 6, 13, 85, 87, 97), \quad (m_1, \dots, m_6) = (1, 1, 1, 1000, 3000, 5). \quad (14.6)$$

In particular $1/100 \leq \rho \leq 1500000 + 1/100$, so $a = 1/\rho$ satisfies all the hypotheses in (14.1).

14.2. A rational certificate on the entire evaluation interval

Here the explicit counterexample is proved by finite exact polynomial arithmetic, not by testing finitely many evaluation points. We give the formulas and a positive Bernstein certificate covering every $x \in [0, 1]$.

Let $A_{ij} = \int_0^1 \rho(t) t^{i+j} dt$, $0 \leq i, j \leq 2$, and let $B_i(x) = \int_0^1 \rho(t) t^i h_x(t) dt$. Then

$$D(x) = \int_0^1 \rho(t) h_x(t)^2 dt - B(x)^T A^{-1} B(x). \quad (14.7)$$

The matrix A is positive definite. All its entries and the required polynomial coefficients are rational. For full reproducibility, define

$$F_{k,r}(x, z) = \sum_{j=0}^r (-1)^j \binom{r}{j} \frac{x^{r-j} z^{k+j+1}}{k+j+1}. \quad (14.8)$$

On a constant-weight interval $[l, s]$ lying to the left of x , its contribution to $\int \rho t^k (x-t)_+^r dt$ is $\rho\{F_{k,r}(x, s) - F_{k,r}(x, l)\}$. On the interval containing x , replace s by x . Intervals to the right contribute zero. Use $(k, r) = (0, 2), (1, 2), (2, 2), (0, 4)$ in (14.7). On each of the thirteen constant-weight intervals, D is thus a polynomial of degree at most ten.

For $Q(x) = 9/10000 - D(x)$, write on a subinterval $[l, s]$

$$Q(l + (s-l)y) = \sum_{k=0}^{10} q_k y^k = \sum_{i=0}^{10} \beta_i \binom{10}{i} y^i (1-y)^{10-i}, \quad \beta_i = \sum_{k=0}^i q_k \frac{\binom{i}{k}}{\binom{10}{k}}. \quad (14.9)$$

A polynomial of lower degree is padded by zero coefficients. The following table gives rational lower bounds: on each row every β_i is at least the displayed integer divided by 10^9 . The only added subdivision of a constant-weight interval is at $49/100$.

1000 <i>l</i>	1000 <i>s</i>	Lower bound numerator
0	29	899999
29	31	899999
31	59	899579
59	61	899452
61	129	863685
129	131	860777
131	490	96932
490	849	88748
849	851	339729
851	869	337888
869	871	480234
871	969	511776
969	971	899999
971	1000	899999

The Bernstein basis is nonnegative and sums to one on $[0, 1]$. Every entry is positive, so the table proves $D(x) < 9/10000$ on the entire closed interval. Exact integration and the same moment projection give

$$J = \frac{8182370250764298579352352052564188484681}{84556305112894848055832725197084050000000000} > \frac{967}{10^6}. \quad (14.10)$$

Equations (14.3)–(14.4) now prove (14.2). The accompanying [certificate generator](#) computes the piecewise polynomials and their exact Bernstein coefficients. A separate [interpolation checker](#) evaluates the integrals at eleven rational points per interval, reconstructs the degree-at-most-ten polynomial, and verifies every printed bound and the exact fraction (14.10). Interpolation is exact here because the degree bound is already proved. Both programs use rational arithmetic for all decisive comparisons. The initial floating-point search is not part of this proof.

14.3. Smooth weights and the limit of freezing

Extend ρ by $1/100$ outside $[0, 1]$ and convolve with nonnegative smooth approximate identities. The resulting ρ_δ are smooth, obey the same positive lower and upper bounds, and converge to ρ in $L^1(0, 1)$. The moment matrices converge to A , so their inverses converge. The kernels h_x are uniformly bounded for $0 \leq x, t \leq 1$; therefore all moments in (14.7), and hence $D_\delta(x)$, converge uniformly in x . The fixed two-switch energy J_δ also converges to J . The strict gaps in (14.2) persist for sufficiently small δ . Taking $a_\delta = 1/\rho_\delta$ proves the smooth assertion. Weighted variation maximizers exist by the compact embedding of the equivalent Hilbert norm into $C^1[0, 1]$. A single-peaked function vanishing at both endpoints has variation twice its height, so none can maximize when $V(a) > 2C(a)$. $\square$

The counterexample closes one tempting route to the original problem. If $1 < p < 2$ and u is a norm-one global maximizer with $f = u^{(n)}$, define $a = |f|^{p-2}$ away from its finite zero set. For every clamped h with finite weighted energy, Holder's inequality gives

$$\int_0^1 |h^{(n)}|^p \leq \left(\int_0^1 a |h^{(n)}|^2 \right)^{p/2} \left(\int_0^1 |f|^p \right)^{(2-p)/2} = \left(\int_0^1 a |h^{(n)}|^2 \right)^{p/2}. \quad (14.11)$$

Thus u also maximizes variation in this weighted Hilbert problem. However, a general weighted single-peak theorem is false by Theorem 15. Any successful use of (14.11) must exploit the special

relation between the weight and the original maximizer. Our example is not shown to have this relation. The corresponding transfer for $p > 2$ is not established by (14.11).

The recent weighted result of Hindov and Lokharu [HL, Theorem 1] places a weight on the function's L^1 norm and leaves the derivative energy unweighted. It has no additional mean constraint of the kind obtained by setting $v = u'$. It therefore supplies neither the arbitrary derivative-weight assertion disproved here nor the missing constrained variation theorem.

15. Nodal-length concentration in order three

In derivative order three, independent changes to the lengths of the nodal intervals of u' give another necessary condition. The changes preserve u'' at the joins, so they remain admissible even though they need not preserve higher derivatives.

Theorem 16 (two dominant nodal intervals). Let $1 < p < \infty$ and let $u \in W_0^{3,p}(0,1)$ be a global variation maximizer, normalized by $\|u'''\|_p = 1$. Let I_j be the finitely many intervals of constant nonzero sign of u' , and put

$$L_j = |I_j|, \quad E_j = \int_{I_j} |u'''|^p. \quad (15.1)$$

At most two intervals satisfy

$$\frac{E_j}{L_j} < \frac{3p-1}{p^2-1}. \quad (15.2)$$

If $1 < p < 3$, there is a set S of at most two intervals such that

$$\sum_{j \in S} L_j > \frac{p(3-p)}{3p-1}, \quad \sum_{j \notin S} L_j < \frac{p^2-1}{3p-1}. \quad (15.3)$$

Thus the total length outside two intervals tends to zero at an explicit rate as $p \downarrow 1$. Additional short intervals are not excluded by this theorem.

15.1. An admissible finite-dimensional family

Put $v = u'$, so $v \in W_0^{2,p}(0,1)$ and $\int_0^1 v = 0$. Theorem 12 supplies finitely many nondegenerate interior zeros of v . Let σ_j be its sign on $I_j = (t_j, t_{j+1})$, and write

$$A_j = \int_{I_j} |v| > 0, \quad M_j = \sigma_j A_j, \quad \sum_j A_j = V, \quad \sum_j M_j = 0, \quad \sum_j E_j = \sum_j L_j = 1, \quad (15.4)$$

where $V = V_{3,p}$. Every E_j is positive: if $v'' = 0$ on I_j , the zero endpoint values would make v identically zero there.

Choose $s_j > 0$ with $\sum_j L_j s_j = 1$, and concatenate intervals of lengths $L_j s_j$ in their original order. If $\tilde{t}_j$ is the new left endpoint, set on the j th interval

$$v_s(x) = s_j v \left(t_j + \frac{x - \tilde{t}_j}{s_j} \right). \quad (15.5)$$

At every join, $v_s = 0$ and the one-sided values of v'_s agree with the same original value of v' . Hence $v_s \in W^{2,p}$ and $v_s = v'_s = 0$ at the outer endpoints. The second derivative on this interval is v''/s_j . If also $\sum_j M_j s_j^2 = 0$, integration gives an admissible $u_s \in W_0^{3,p}$.

The functionals are exactly

$$\begin{aligned} T(s) &= \sum_j A_j s_j^2, & E(s) &= \sum_j E_j s_j^{1-p}, \\ M(s) &= \sum_j M_j s_j^2, & L(s) &= \sum_j L_j s_j, \end{aligned} \quad F(s) = T(s)E(s)^{-1/p}. \quad (15.6)$$

The point $s = \mathbf{1}$ is a local maximum of F on $L(s) = 1$, $M(s) = 0$. These two constraints have independent gradients: the vector (L_j) is strictly positive, while (M_j) has both signs. Their tangent space is therefore

$$\sum_j L_j x_j = 0, \quad \sum_j M_j x_j = 0. \quad (15.7)$$

Only the explicit smooth finite-dimensional functionals in (15.6) are differentiated. Differentiability of $s \mapsto u_s$ in a Sobolev norm is not needed.

15.2. The constrained second variation

Put $q = p/(p-1)$ and $d = 2q + 1$. Lagrange multipliers η, ν give

$$2A_j + \frac{V}{q}E_j - 2\eta M_j - \nu L_j = 0, \quad \nu = \frac{d}{q}V, \quad 2(A_j - \eta M_j) = \frac{V}{q}(dL_j - E_j). \quad (15.8)$$

The value of ν follows by summing and using (15.4). On the tangent space (15.7), the same equation yields

$$\sum_j A_j x_j = -\frac{V}{2q} \sum_j E_j x_j. \quad (15.9)$$

The Hessian of $F - \eta M - \nu(L - 1)$, restricted to that tangent space, is

$$Q(x) = \frac{V}{q} \left\{ \sum_j [dL_j - (p+1)E_j] x_j^2 + \frac{p-1}{q} \left(\sum_j E_j x_j \right)^2 \right\} \leq 0. \quad (15.10)$$

Before applying (15.9), the two contributions to the rank-one term are

$$\frac{4}{q} \left(\sum_j A_j x_j \right) \left(\sum_j E_j x_j \right) \quad \text{and} \quad \frac{V(p+1)}{q^2} \left(\sum_j E_j x_j \right)^2.$$

Their sum becomes $V(p-1)(\sum_j E_j x_j)^2/q^2$. The diagonal part is

$$\sum_j [2(A_j - \eta M_j) - V(p-1)E_j] x_j^2.$$

Substituting (15.8) gives (15.10).

If three indices had $dL_j - (p+1)E_j > 0$, a nonzero vector supported on those three indices could satisfy the two equations (15.7). Its diagonal contribution in (15.10) would be strictly positive, and the rank-one contribution is nonnegative. This contradicts $Q(x) \leq 0$ and proves (15.2), since $d/(p+1) = (3p-1)/(p^2-1)$.

For $1 < p < 3$, let S consist of the indices in (15.2). It is nonempty, because

$$\sum_j [dL_j - (p+1)E_j] = d - (p+1) > 0. \quad (15.11)$$

It contains at most two indices. On its complement, $E_j \geq dL_j/(p+1)$, so

$$\frac{d}{p+1} \sum_{j \notin S} L_j \leq \sum_{j \notin S} E_j < 1. \quad (15.12)$$

The strict inequality uses the positive energy on the nonempty set S . Since $1 - (p+1)/d = p(3-p)/(3p-1)$, this proves (15.3). $\square$

Theorem 16 gives a quantified restriction on the geometry of any remaining maximizer in order three. For example, the two selected intervals cover more than $9/14$ of the domain at $p = 3/2$, and more than $2/5$ at $p = 2$. It does not bound the number of additional short intervals. The direct rescaling argument does not extend to higher derivative orders, because preserving v' at the joins does not preserve all the additional derivatives then required.

16. The complete finite-exponent theorem in order three

We now settle the variation identity and the full equality classification in derivative order three at every finite exponent.

Theorem 17 (order three at every finite exponent). For every $1 < p < \infty$,

$$V_{3,p} = 2C_{3,p}. \quad (16.1)$$

The nonzero variation extremizers are exactly the scalar multiples of the unique positive norm-one midpoint height extremizer. They are symmetric about the midpoint and have exactly one interior critical point. In particular, [NS, Conjecture 4.14] holds for $n = 3$ and every finite $p > 1$.

The proof retains the residual equation, which contains more information than the interval-energy conditions in Section 15. It reconstructs every possible global maximizer from its residual vertices, then compares a positive action across different nodal counts. The comparison is a hand proof for every count and exponent.

16.1. Normalization and the conserved quantity

Let $q = p/(p-1) > 1$. Use the unnormalized dual potential U and set $v = U'$, so

$$v'' = \phi_q(R), \quad \phi_q(s) = |s|^{q-2}s, \quad R'' = \text{sign}(v) - c.$$

The endpoint conditions are $v = v' = 0$ at both endpoints and $\int v = 0$. For a global maximizer, Theorem 12 supplies finitely many simple interior zeros of v , and $-1 < c < 1$. Its dual residual is continuously differentiable and quadratic between those zeros. Put $J = \int |R|^q$.

On any interval of constant sign of v , direct differentiation gives

$$\mathcal{H} = R'v' - R''v - \frac{|R|^q}{q}, \quad \mathcal{H}' = 0. \quad (16.2)$$

This remains valid for $1 < q < 2$: the derivative of $|R|^q/q$ is the continuous function $\phi_q(R)R'$. No derivative of ϕ_q is needed. At a switch, R'' jumps but $v = 0$, so $\mathcal{H}$ has no jump. The endpoint conditions imply

$$\mathcal{H} = -\frac{A^q}{q}, \quad A = |R(0)| = |R(1)| > 0. \quad (16.3)$$

Integrating (16.2) on an interval of total length ℓ , and integrating by parts using $v = v' = 0$, gives

$$\int R'v' = -J, \quad \int R''v = J, \quad A^q\ell = (2q+1)J. \quad (16.4)$$

When $\ell = 1$, this recovers the published endpoint normalization. The identity (16.4) uses the two endpoint conditions on v ; the extra mean condition is not needed for this particular identity.

16.2. Strict interior bound and one vertex in every nodal interval

Lemma. Every interior point satisfies $|R(t)| < A$. In every nodal interval of v there is exactly one zero of R' , and the sign of R there is opposite to the sign of v .

At an interior maximum of $|R|$, one has $R' = 0$. If this point is not a switch, then $R''v = (1 - c \operatorname{sign} v)|v| > 0$. Equations (16.2)-(16.3) imply $|R|^q < A^q$. If it is a switch, they imply $|R| = A$. Thus the maximum over the closed interval is at most A .

An interior equality point must therefore be a switch t_0 with $R'(t_0) = 0$. The two nonzero constant curvatures on its sides have opposite signs. Since R' vanishes at the switch, it has the same strict sign immediately on both sides. Hence R passes strictly through its value there, which contradicts the global bound $|R| \leq A$. This proves strictness.

At an interior switch, (16.2)-(16.3) now give

$$R'v' = \frac{|R|^q - A^q}{q} < 0. \quad (16.5)$$

Let σ be the sign of v on a nodal interval and $a = 1 - c\sigma > 0$. The derivative R' is affine with slope σa . At an interior left endpoint it has sign $-\sigma$, and at an interior right endpoint it has sign σ , by (16.5) and the simple crossing signs of v' .

At the outer left endpoint, $R(0) = \sigma A$: this follows from $v(0) = v'(0) = 0$ and $v''(0) = \phi_q(R(0))$. Strict decrease of σR into the interval gives $\sigma R'(0) < 0$; equality would contradict the positive curvature of σR . The corresponding right endpoint argument gives $\sigma R'(1) > 0$. Consequently every nodal interval has exactly one interior vertex of R .

Finally, on a nodal interval I ,

$$\int_I v\phi_q(R) = \int_I vv'' = -\int_I (v')^2 < 0.$$

Thus σR is negative somewhere. Its unique minimum is therefore $-h$ for some $0 < h < A$. This proves the lemma.

For the normalized original maximizer, the lemma also gives $|u'''(t)|^p < 2q + 1$ at every interior point. This is stronger than a bound only at the endpoints but does not by itself force the average energy-density threshold in Theorem 16.

16.3. Matching consecutive quadratic pieces

Let the N nodal intervals have alternating signs $\sigma_j = (-1)^{j-1}$ after a global sign choice, and set $a_j = 1 - c\sigma_j$. Write the residual on interval j , in coordinates y centered at its vertex, as

$$R = \sigma_j P_j(y), \quad P_j(y) = \frac{a_j y^2}{2} - h_j, \quad -\alpha_j \leq y \leq \beta_j. \quad (16.6)$$

At the join between j and $j + 1$, continuity of R' and R gives

$$s_j = a_j \beta_j = a_{j+1} \alpha_{j+1}, \quad s_j^2 = a_j a_{j+1} (h_j + h_{j+1}). \quad (16.7)$$

Indeed, $a_j + a_{j+1} = 2$, and substitution of the matching slopes into the matching values yields (16.7). The outer endpoint values give

$$\alpha_1 = \sqrt{\frac{2(A + h_1)}{a_1}}, \quad \beta_N = \sqrt{\frac{2(A + h_N)}{a_N}}. \quad (16.8)$$

Thus all half-lengths are determined by A , c , and the positive vertex depths $h_j < A$.

16.4. The exact nodal equation

On interval j , put $w = \sigma_j v > 0$. At its vertex, (16.2) gives

$$b_j = w(0) = \frac{A^q - h_j^q}{q a_j} > 0.$$

Since $w'' = \phi_q(P_j)$ and P_j is even, for some real tilt B_j one has

$$w(y) = b_j + B_j y + I_j(|y|), \quad I_j(t) = \int_0^t (t - z) \phi_q(P_j(z)) dz. \quad (16.9)$$

Define

$$\Psi_j(t) = \frac{b_j + I_j(t)}{t}.$$

The two endpoint equations $w(-\alpha_j) = w(\beta_j) = 0$ are equivalent to

$$\Psi_j(\alpha_j) + \Psi_j(\beta_j) = 0, \quad B_j = \Psi_j(\alpha_j). \quad (16.10)$$

A useful derivative identity is

$$\Psi'_j(t) = \frac{|P_j(t)|^q - A^q}{qa_j t^2}. \quad (16.11)$$

To check it, use $P'_j = a_j t$ to integrate $z\phi_q(P_j(z))$. The h_j^q terms cancel against b_j . In particular, Ψ_j is strictly decreasing wherever $|P_j| < A$.

16.5. Sufficiency and the remaining mean condition

Conversely, take $q > 1$, $A > 0$, $-1 < c < 1$, and $0 < h_j < A$, with half-lengths defined by (16.7)-(16.8). Suppose all N equations (16.10) hold. At every internal join the residual value is

$$\sigma_j R = \frac{a_{j+1}h_{j+1} - a_j h_j}{2},$$

which lies strictly between $-A$ and A . The quadratic pieces themselves remain in this range away from the outer endpoints. Hence (16.11) proves $w > 0$ between its two zeros: on the right,

$$w(t) = t[\Psi_j(t) - \Psi_j(\beta_j)] > 0,$$

and the left half is identical with α_j in place of β_j .

The pieces (16.9) glue to a continuously differentiable v . In fact, at either internal join their one-sided derivatives both satisfy (16.5), with the same nonzero R' . Thus these derivatives agree. The outer derivatives vanish because $|R| = A$ there. Since $v'' = \phi_q(R)$ is continuous, the resulting v is C^2 and has the required simple interior zeros.

Set

$$L_j = \alpha_j + \beta_j, \quad \mathcal{E}_j = \int_{-\alpha_j}^{\beta_j} |P_j(y)|^q dy.$$

Integrating the conserved quantity on a nodal interval gives its positive area

$$\mathcal{A}_j = \int_{I_j} |v| = \frac{A^q L_j - \mathcal{E}_j}{2qa_j}. \quad (16.12)$$

Consequently the last condition needed for $U = \int v$ to have all three zero endpoint traces is precisely

$$\sum_j \sigma_j \mathcal{A}_j = 0. \quad (16.13)$$

Under (16.13), $\phi_q(R) = U'''$ annihilates every polynomial of degree at most two by integration by parts. Also $R'' = \text{sign}(v) - c$, so R differs from the integrated sign load by a quadratic. These are exactly the normal equations for its unique best L^q approximation by quadratics. Thus the constructed profile is a genuine sign-compatible stationary dual configuration, not merely an abstract collection of lengths and energies. This converse does not assert local or global maximality.

16.6. Removing the overall scale and identifying the comparison target

Normalize the endpoint residual magnitude to $A = 1$, and solve (16.10) and (16.13) for the $N + 1$ unknowns $h_1, \dots, h_N, c$, with $0 < h_j < 1$. Let

$$\Lambda = \sum_j (\alpha_j + \beta_j), \quad d = 2q + 1.$$

The unit-interval residual is obtained by multiplying lengths by Λ^{-1} and residual values by Λ^{-2} . Its energy and variation quotient are

$$J_{\text{unit}} = \frac{\Lambda^{-2q}}{d}, \quad \mathcal{V}_{\text{unit}} = J_{\text{unit}}^{1/q} = \Lambda^{-2} d^{-1/q}. \quad (16.14)$$

Indeed (16.4) on the normalized interval gives $J = \Lambda/d$, and the rescaling factor in its energy is $\Lambda^{-(2q+1)}$. Also $\int |v| = J$ follows from (16.13), so Holder equality identifies the quotient.

Every global maximizer is covered by this finite system by Theorem 12. The next argument constructs a better admissible competitor whenever $N \geq 4$. Published Theorem 13 already excludes $N = 3$.

At $q = 2$, the single-switch solution has $c = 0$ and $h_1 = h_2 = 9/16$. Its half-lengths are $5/(2\sqrt{2})$ and $3/(2\sqrt{2})$ in each interval, giving $\Lambda^2 = 32$ and $\mathcal{V} = 1/(32\sqrt{5})$, in agreement with Theorem 1. The all- q single-switch profile is supplied by Theorem 10.

16.7. A positive action for matched intervals

Fix $q > 1$, $N \geq 2$, alternating signs $\sigma_j = (-1)^{j-1}$, depths $h \in (0, 1)^N$, and $c \in (-1, 1)$. Put $a_j = 1 - c\sigma_j$ and $f(r) = 1 - |r|^q$. Define the positive outer contribution

$$O(h, a) = \frac{1}{\sqrt{2a}} \int_{-h}^1 \frac{f(r)}{\sqrt{r+h}} dr. \quad (16.15)$$

For consecutive depths h, k and curvatures a, b with $a + b = 2$, define

$$B(h, k; a, b) = \max_{-h \leq z \leq k} \left\{ \frac{1}{\sqrt{2a}} \int_{-h}^z \frac{f(r)}{\sqrt{r+h}} dr + \frac{1}{\sqrt{2b}} \int_{-k}^{-z} \frac{f(r)}{\sqrt{r+k}} dr \right\}. \quad (16.16)$$

The maximizing join is unique and interior. Indeed the derivative in z is

$$f(z) \left[\frac{1}{\sqrt{2a(z+h)}} - \frac{1}{\sqrt{2b(k-z)}} \right].$$

Its bracket is strictly decreasing from positive infinity to negative infinity. Since $f(z) > 0$ on $[-h, k]$, its unique zero is

$$z = \frac{bk - ah}{2}. \quad (16.17)$$

This is precisely the matching join in (16.7). Consequently

$$K_N(h, c) = O(h_1, a_1) + O(h_N, a_N) + \sum_{j=1}^{N-1} B(h_j, h_{j+1}; a_j, a_{j+1}) = \sum_j \int_{-\alpha_j}^{\beta_j} f(P_j(y)) dy. \quad (16.18)$$

Every contribution is positive.

16.8. The depths maximize a strictly concave function

For fixed c , direct differentiation of the last expression in (16.18) gives

$$\frac{\partial K_N}{\partial h_j} = q[\Psi_j(\alpha_j) + \Psi_j(\beta_j)]. \quad (16.19)$$

Here is the endpoint accounting. An internal join has slope magnitude $s = \sqrt{ab(h+k)}$. Varying h changes the sum of its two half-lengths by $1/s$. The two matching residual values have the same absolute value, so their endpoint contribution is $f(z)/s$. The integral contribution on the changed interval is $q \int \phi_q(P_j)$. Since $s = a_j t$, these are exactly $q\Psi_j(t)$ in the alternative identity

$$\Psi_j(t) = \int_0^t \phi_q(P_j(y)) dy + \frac{1 - |P_j(t)|^q}{qa_j t}.$$

At an outer endpoint $P_j = 1$, the endpoint contribution is zero and this same formula applies.

For fixed a and t , differentiating the original definition $\Psi = (b + I)/t$ in h gives

$$\Psi_h = -\frac{h^{q-1}}{at} - \frac{q-1}{t} \int_0^t (t-y)|ay^2/2 - h|^{q-2} dy < 0. \quad (16.20)$$

The integral is finite for all $q > 1$: its possible zero is simple, because $h > 0$. At a zero located exactly at t , the factor $t - y$ also removes any endpoint concern. The derivative follows by splitting at the moving simple zero, or by local power estimates. Thus the depth gradient is continuously differentiable in the open cube.

The derivative Ψ_t is negative at an internal half-length by (16.11). That half-length has equal positive derivatives with respect to its own depth and its neighboring depth. The outer half-length has positive own-depth derivative and $\Psi_t = 0$ there. Therefore the Hessian of K_N is symmetric, has negative diagonal entries and nonpositive off-diagonal entries, and is strictly diagonally dominant: the two strictly negative fixed- t terms (16.20) supply the strict excess on each diagonal. Hence it is negative definite. So $K_N(\cdot, c)$ is strictly concave.

A maximum exists in the interior. The function extends continuously to $[0, 1]^N$ by its integral definitions; zero-length internal pieces contribute zero. If h_j is sufficiently close to zero, uniformly over the other depths, both relevant Ψ terms are positive. One bound is

$$\Psi(t) \geq \frac{1 - h_j^q}{qa_j t} - \frac{h_j^{q-1} t}{2},$$

and the half-lengths have a fixed upper bound when c is fixed. If h_j is sufficiently close to one, both terms are negative uniformly over the other depths. At $h_j = 1$, $b = 0$, $\Psi(0+) = 0$, and (16.11) is strictly negative up to the outer endpoint. All half-lengths have a uniform positive lower bound when h_j is near one. Compactness and continuity give the asserted uniform negativity. Thus every maximizing depth lies in an interior cube. Strict concavity gives a unique maximizing depth vector $h(c)$, and (16.19) says it solves exactly the N nodal equations. Conversely every solution of those equations is this maximizer.

For a second uniqueness argument not requiring (16.20), choose an index with maximal absolute difference between two proposed solutions. After exchanging the solutions, its own depth increases and the sum with either neighboring depth does not decrease. Hence both half-lengths do not decrease. Each Ψ decreases strictly with its own depth and decreases with the half-length throughout

the admissible range. Its nodal equation cannot remain zero. This contradicts the existence of two solutions.

16.9. Strict convexity in the mean parameter

For fixed h, k, z , each integral in (16.16) is nonnegative and independent of c ; its coefficient is a positive multiple of $(1 - c\sigma)^{-1/2}$, a strictly convex function of c . Thus B is convex in c , being the maximum over a fixed interval of convex functions. The outer terms are strictly convex and positive. Therefore $K_N(h, \cdot)$ is strictly convex.

Set $G_N(c) = \max_h K_N(h, c) = K_N(h(c), c)$. This is convex as a supremum of convex functions, and is strictly convex because the positive outer terms supply strictness. More explicitly, for two distinct parameter values take the maximizing h at their strict convex combination, apply strict convexity for that same h , and then bound its endpoint values by the corresponding maxima.

Moreover $G_N(c) \rightarrow \infty$ as $c \rightarrow \pm 1$. For $c \rightarrow 1$, the positive first outer interval suffices. For $c \rightarrow -1$, an even N has a negative last outer interval. For odd $N \geq 3$, choose fixed positive depths and, in an internal edge adjacent to a negative interval, a fixed interior join such as zero; its positive integral has coefficient $(1 + c)^{-1/2}$. The definition as a maximum gives divergence. Thus G_N has a unique minimum at some $c_N \in (-1, 1)$.

The nondegenerate depth Hessian gives differentiable dependence on c by the implicit function theorem. Differentiating (16.16) at its optimizing join and then (16.18) gives

$$G'_N(c) = \frac{\partial K_N}{\partial c}(h(c), c) = \sum_j \frac{\sigma_j}{2a_j}(L_j - \mathcal{E}_j) = q \sum_j \sigma_j \mathcal{A}_j. \quad (16.21)$$

The moving-join term vanishes because (16.17) is its critical point; the moving-depth terms vanish by (16.19). Hence c_N is exactly the unique parameter satisfying the zero-mean condition. The shooting reconstruction proves that this minimum supplies an actual admissible sign-compatible stationary profile on rescaling to the unit interval.

For every c , reconstruction gives the two zero endpoint traces of v , whether or not its mean is zero. Therefore (16.4) yields

$$G_N(c) = \frac{2q}{2q + 1} \Lambda_N(c). \quad (16.22)$$

A stationary mean-zero profile with N intervals thus has normalized total length $\Lambda_N(c_N)$ and variation quotient given by (16.14).

16.10. Removing a pair of intervals improves the quotient

For $N \geq 4$ and a fixed c , take the maximizing depth vector of K_{N-2} . Append two copies of its final depth. The final sign is unchanged by adding two intervals, so the two outer contributions and every old edge contribution are unchanged. Exactly two additional edge contributions appear, and both are strictly positive. Thus

$$G_N(c) > G_{N-2}(c). \quad (16.23)$$

Evaluate this at c_N and use the minimum defining c_{N-2} :

$$G_N(c_N) > G_{N-2}(c_N) \geq G_{N-2}(c_{N-2}). \quad (16.24)$$

By (16.22) and (16.14), the admissible stationary profile with $N - 2$ intervals has a strictly larger variation quotient than the profile with N intervals. Therefore no global variation maximizer can have $N \geq 4$ nodal intervals.

A global maximizer cannot have $N = 1$ because its nonzero first derivative has zero mean. The case $N = 3$ is excluded by published Theorem 13 (two interior critical points). Thus $N = 2$ for every global maximizer in derivative order three and every finite $p > 1$. Such a function has a single peak, so its variation is twice its height. Published Theorem 10 supplies the unique midpoint-symmetric height extremizer, up to scalar multiplication. This establishes

$$V_{3,p} = 2C_{3,p}, \quad 1 < p < \infty,$$

with coincidence and midpoint symmetry of the extremizers. This proves Theorem 17. $\square$

16.11. The constant at the uniform endpoint

Corollary (uniform norm constant). One also has

$$V_{3,\infty} = 2C_{3,\infty}. \quad (16.25)$$

Proof. The height constants $C_{3,p}$ decrease as p increases and are bounded below by $C_{3,\infty}$. To identify their limit, take normalized height maximizers with $p \rightarrow \infty$. For every fixed $s > 1$, their third derivatives eventually have L^s norm at most one. Successive integration and compactness give a subsequence converging through the second derivative. Weak compactness in every fixed L^s , followed by a diagonal extraction, gives a limit whose third derivative has L^s norm at most one for every finite s . On an interval of length one this implies its L^∞ norm is at most one: any set on which its magnitude exceeded $1 + \varepsilon$ would contradict the bound for sufficiently large s . Thus the limiting height is at most $C_{3,\infty}$, and $C_{3,p} \rightarrow C_{3,\infty}$.

For any clamped u with $\|u'''\|_\infty \leq 1$, Theorem 17 gives $\|u'\|_1 \leq 2C_{3,p}$ for every finite $p > 1$. Passing to the limit proves the upper bound in (16.25); the reverse bound follows from variation being at least twice the height. $\square$

The corollary proves the endpoint constant identity. Section 17 establishes the complete variation equality classification at this endpoint in order three. The measure endpoint and its equality classification were already settled by Theorem 9. For derivative orders at least four, the remaining finite-exponent variation problem is still open.

17. Complete equality at the uniform endpoint

Theorem 18 (uniform endpoint in order three). One has

$$V_{3,\infty} = 2C_{3,\infty} = \frac{2 - \sqrt{2}}{48}. \quad (17.1)$$

Every nonzero variation extremizer is a scalar multiple of the following function. Put $a = (1 - 1/\sqrt{2})/2$, let f_* take the successive values $+1, -1, +1, -1$ on the intervals cut out by $0, a, 1/2, 1 - a, 1$, and define

$$u_*(x) = \frac{1}{2} \int_0^x (x-t)^2 f_*(t) dt. \quad (17.2)$$

The height and variation extremizers coincide and are symmetric about the midpoint. The height constant is credited prior work [NS, Theorems 4.5 and 4.12]; the new step here is the classification of every variation maximizer. Together with Theorems 9 and 17, this completes [NS, Conjecture 4.14] in derivative order three for all exponents, with the measure interpretation at $p = 1$.

17.1. Duality for an arbitrary endpoint maximizer

The unit ball of clamped $W^{3,\infty}$ is compact through the second derivative and weak-star closed in the third derivative, so the variation supremum is attained. Let u be any maximizer with $\|u'''\|_\infty = 1$, and set $v = u'$, $f = u'''$, and $g = \text{sign } v$, with sign zero defined to be zero. The third derivative annihilates every quadratic polynomial.

The quotient-space Hahn-Banach formula identifies the norm of the linear load g on this annihilator with the distance of its integrated load H_g from the quadratics in L^1 . A best quadratic exists by finite-dimensional coercivity. For any such best quadratic P , put $R = H_g - P$. Since g supports this variation maximizer, equality in the dual pairing gives

$$J := \int_0^1 |R| = \int_0^1 |v| > 0, \quad f = \text{sign } R \quad \text{a.e. on } \{R \neq 0\}, \quad R'' = g - c \quad (17.3)$$

for a real constant c . On the zero set of R , only $|f| \leq 1$ is initially known. No finite-switch or bang-bang assumption has been made. The pairing equality follows explicitly from $\int Rf = \int H_g f = \int gv$ and the dual norm formula.

17.2. The mean parameter is strictly between minus and plus one

Suppose $c \geq 1$. Then R is concave. If it is positive somewhere, its positive set is an interval, and its zeros outside that set have measure zero. On the negative pieces reaching the outer endpoints, $v'' = -1$, hence $v = -t^2/2$ on the left piece and $v = -(1-t)^2/2$ on the right piece. On the positive interval, $v'' = 1$ and v is convex with nonpositive endpoint values. Thus $v \leq 0$ everywhere, contradicting its zero mean and nontriviality. This argument also covers a positive interval touching an outer endpoint, where the corresponding endpoint value of v is zero. A concave function positive somewhere cannot have a separate zero interval, since its derivative is nonincreasing.

If instead $R \leq 0$ everywhere, its zero set is an interval, a point, or empty. If that set has measure zero, $v'' = -1$ almost everywhere, contradicting the two zero endpoint values of v' . Otherwise write the zero interval as $[\alpha, \beta]$. There $R'' = 0$, so $g = c$ almost everywhere. Necessarily $c = 1$ and $v > 0$ almost everywhere on its interior. But continuity at the left end gives $v(\alpha) = -\alpha^2/2$, forcing $\alpha = 0$; the right end similarly forces $\beta = 1$. This would make R identically zero, contradicting $J > 0$. The case $c \leq -1$ follows by changing the signs of u, R, c . Therefore

$$-1 < c < 1. \quad (17.4)$$

17.3. The conserved quantity proves endpoint regularity

The absolutely continuous function

$$\mathcal{H} = R'v' - |v| + cv - |R| \quad (17.5)$$

has derivative $R'(f - \text{sign } R) = 0$ almost everywhere. On the zero set of R , this uses the fact that an absolutely continuous function has derivative zero almost everywhere on a level set. Thus the argument does not differentiate the discontinuous sign function and does not assume that the zero set is finite.

The endpoint traces and an integration by parts give

$$\mathcal{H} = -A, \quad A = |R(0)| = |R(1)| = 3J > 0. \quad (17.6)$$

Indeed $\int R'v' = -J$, $\int(|v| - cv) = J$, and $\int |R| = J$.

At an interior maximum of $|R|$, $R' = 0$, so (17.4)-(17.6) show $|R| \leq A$. Thus $|R| \leq A$ everywhere. If an interior zero of v also had $v' = 0$, (17.5) would give $|R| = A$, and hence $R' = 0$ there. In a small neighborhood, R has one nonzero sign ε , so $v'' = \varepsilon$ and $v(t) = \varepsilon(t - t_0)^2/2$. Then $R'' = \varepsilon - c$ and

$$R(t) = \varepsilon A + \frac{\varepsilon - c}{2}(t - t_0)^2,$$

contradicting $|R| \leq A$. Every interior zero of v is therefore simple. The nonzero endpoint residuals also give a fixed quadratic sign of v near each outer endpoint. Compactness now rules out infinitely many interior zeros: an accumulation point would have both $v = v' = 0$.

Finally, equality $|R| = A$ at an interior point forces $v = 0$. At this simple zero, $R' = 0$ would combine with the two opposite nonzero curvatures to make R pass monotonically through that value, again contradicting its bound. Consequently

$$|R(t)| < A \quad (0 < t < 1), \quad R'v' = |R| - A < 0 \quad \text{at every interior zero of } v. \quad (17.7)$$

Exactly as in Section 16, each of the finitely many nodal intervals of v has one residual vertex, of sign opposite to v and magnitude $h_j \in (0, A)$. The opposite sign also follows from $\int_I vv'' = -\int_I (v')^2 < 0$. The alternating vertices and the two endpoint values imply exactly $N + 1$ simple residual roots when there are N nodal intervals. Thus $f = \text{sign } R$ almost everywhere on the whole domain, and v is $C^{1,1}$. These conclusions were derived for an arbitrary endpoint maximizer.

17.4. The positive action extends to $q=1$

Normalize $A = 1$. Keep the half-lengths, outer action, edge action and total action of Section 16, with the integrand $1 - |r|$ in place of $1 - |r|^q$. On a nodal interval write $P(y) = ay^2/2 - h$, where $a = 1 - c\sigma > 0$, $0 < h < 1$, and put $r_0 = \sqrt{2h/a}$. The exact scalar function is now

$$\Psi(t) = \frac{1-h}{at} - \frac{t}{2} + \frac{(t-r_0)_+^2}{t}, \quad \Psi_t = \frac{|P(t)|-1}{at^2}, \quad \Psi_h = -\frac{1}{at} - \frac{2(t-r_0)_+}{ar_0t} < 0. \quad (17.8)$$

In particular, Ψ is continuously differentiable even when a residual root meets a nodal endpoint. Simply substituting $q = 1$ into (16.20) would omit the second term in Ψ_h ; formula (17.8) is the needed endpoint calculation.

The action gradient is still the sum of the two Ψ values. Differentiation through $|P|$ is justified by its Lipschitz bound and its simple zeros; (17.8) then gives the continuous second derivatives. The

Hessian is negative definite by the same strict diagonal-dominance argument. Its off-diagonal entries come only from the matched half-lengths, whose own-depth and neighboring-depth derivatives agree; the strictly negative fixed-half-length derivative supplies the excess on the diagonal.

The depth maximum is again interior. To check the zero-depth face, let t be either half-length, bounded above for fixed c . If $t < r_0$, then

$$\Psi(t) \geq \frac{1-2h}{ar_0} > 0 \quad (h < 1/2).$$

If $t \geq r_0$, then

$$\Psi(t) = \frac{1+h}{at} + \frac{t}{2} - 2r_0 \geq \frac{1}{at} - 2r_0 > 0$$

uniformly for sufficiently small h . The upper-depth face follows from $\Psi(0+) = 0$ at $h = 1$, strict decrease in t , and the uniform positive lower bound on the half-lengths. This proves existence and uniqueness of the interior depth maximizer for each c , just as before.

The edge action remains a maximum over a fixed joining-value interval of positive integrals times $(1 - c\sigma)^{-1/2}$. Therefore the optimized action $G_N(c)$ is strictly convex and coercive for every $N \geq 2$. Its derivative is the signed area. Its unique minimum gives a reconstructed clamped $W^{3,\infty}$ profile with zero mean of its first derivative and third derivative equal to the sign of its residual almost everywhere. The reconstruction matches the value and first derivative of v at every join; bounded second derivatives suffice here, and continuity of those second derivatives is not claimed.

The normalized total length Λ_N and quotient are

$$G_N = \frac{2}{3}\Lambda_N, \quad \mathcal{V}_N = \frac{1}{3\Lambda_N^2}. \quad (17.9)$$

All ingredients of the two-interval insertion comparison remain valid. Thus for $N \geq 4$, minimizing after appending two copies of the last depth yields

$$G_N(c_N) > G_{N-2}(c_{N-2}), \quad (17.10)$$

and an actual admissible profile with $N - 2$ intervals has a strictly larger variation. No endpoint global maximizer has $N \geq 4$.

17.5. The three-interval profile is strictly worse

For $N = 3$, reflection preserves the alternating sign sequence $+, -, +$ and the same parameter c . The unique depth vector at that parameter is therefore reflection symmetric. Its residual and its third derivative f are symmetric about the midpoint. The four simple residual roots have the form $b, d, 1 - d, 1 - b$, and f has successive signs $+, -, +, -, +$.

The zero moments of f give

$$d - b = \frac{1}{4}, \quad \int_b^d (t - 1/2)^2 dt = \frac{1}{48}, \quad b = \frac{3 - \sqrt{5}}{8}, \quad d = b + \frac{1}{4}. \quad (17.11)$$

The first zero of v is $\tau = (2 + \sqrt{2})b$, which lies between b and d . On this interval,

$$v(t) = \frac{t^2}{2} - (t-b)^2, \quad u(t) = \frac{t^3}{6} - \frac{(t-b)^3}{3}.$$

The symmetry of v and its zero mean make u antisymmetric. Its variation is therefore

$$\mathcal{V}_3 = 4u(\tau) = 4 \left(1 + \frac{2\sqrt{2}}{3} \right) b^3 = \frac{(3 + 2\sqrt{2})(9 - 4\sqrt{5})}{48}. \quad (17.12)$$

This is strictly below the value in (17.1) of the admissible two-interval profile verified below. An elementary rational comparison suffices: $b < 1/10$ because $\sqrt{5} > 11/5$, and $1 + 2\sqrt{2}/3 < 2$ because $\sqrt{2} < 3/2$. Hence $\mathcal{V}_3 < 1/125 < 1/96 < (2 - \sqrt{2})/48$. The same inequalities show $\tau < d$. Thus $N = 3$ is excluded without assuming the finite-exponent two-switch argument continues to the endpoint.

17.6. The two-interval profile and complete equality

Only $N = 2$ remains, since $N = 1$ contradicts zero mean. Reflection exchanges the signs and sends c to $-c$, so G_2 is even. Its unique minimum is at zero, and the unique depth vector has equal entries. The residual is antisymmetric and has three simple roots $a, 1/2, 1 - a$. The third derivative of the maximizing function has the four signs in (17.2).

The first moment condition, after using antisymmetry, is

$$a^2 - a + \frac{1}{8} = 0, \quad a = \frac{1 - 1/\sqrt{2}}{2}. \quad (17.13)$$

All three moments vanish for this control. On the first half, u'_* is positive until its zero at $1/2 = a(2 + \sqrt{2})$; the second half follows by reflection. Direct integration gives

$$u_*(1/2) = \frac{2 - \sqrt{2}}{96}, \quad \|u'_*\|_1 = 2u_*(1/2) = \frac{2 - \sqrt{2}}{48}. \quad (17.14)$$

Thus every normalized variation maximizer is u_* or $-u_*$. Its height is half its variation. Conversely, any height maximizer has variation at least twice that height and at most $V_{3,\infty}$, so it is a variation maximizer as well. This proves the complete equality classification and the endpoint constant formula. The scalar multiples include all nonzero extremizers; the zero function gives the trivial equality.

18. Uniform-endpoint regularity in every order

The conserved quantity extends to all derivative orders if its additional mixed derivative terms are retained. It proves saturation and finite simple switching for every endpoint maximizer, even though it does not supply the quadratic comparison used in order three.

Theorem 19 (all-order endpoint regularity). Let $n \geq 2$, and let u be any maximizer of variation in the clamped class with $\|u^{(n)}\|_\infty \leq 1$. Put $m = n - 1$, $v = u'$, $f = u^{(n)}$, and $g = \text{sign } v$, with $\text{sign } 0 = 0$. For any best L^1 polynomial approximant $P \in \mathcal{P}_m$ to H_g , put $R = H_g - P$. Then

$$R^{(m)} = (-1)^m(g - c), \quad -1 < c < 1, \quad (18.1)$$

for a real constant c . With $J = V_{n,\infty} > 0$, one has

$$\int_0^1 |R| = \int_0^1 Rf = \int_0^1 |v| = J, \quad |R(0)| = |R(1)| = nJ. \quad (18.2)$$

Both R and v have finitely many simple interior zeros. If v has r such zeros, then $r \geq 1$ and R has exactly $n + r - 1$. In particular,

$$f = \text{sign } R \quad \text{almost everywhere}, \quad |u^{(n)}| = 1 \quad \text{almost everywhere.} \quad (18.3)$$

The highest derivative is constant almost everywhere on a neighborhood of either endpoint, and v is a nonzero signed monomial on each such neighborhood. For each fixed $n \geq 3$, the critical-point count is uniformly bounded over all $1 < p \leq \infty$, as proved below.

18.1. Duality and the conserved quantity

Existence follows from weak-star compactness of the L^∞ unit ball under its n moment constraints, together with uniform convergence of the integrated first derivative. A nonzero smooth clamped function shows that $J > 0$, and homogeneity makes the highest-derivative norm of every maximizer equal to one.

The annihilator of $\mathcal{P}_m$ in L^∞ is exactly the set of clamped n th derivatives. The L^1 quotient norm of H_g modulo $\mathcal{P}_m$ is the norm of the load g on this annihilator. It is at most J , while its pairing with the present f is J . A best polynomial exists by finite-dimensional coercivity. This proves the integral equalities in (18.2), and equality in $Rf \leq |R|$ gives $f = \text{sign } R$ off the residual zero set. On that zero set only $|f| \leq 1$ is initially known.

Both R and v belong to $W^{m,\infty}$, and differentiating the integrated load gives the first identity in (18.1). Define the absolutely continuous function

$$\mathcal{H} = \sum_{i=1}^{m-1} (-1)^{m-1-i} R^{(m-i)} v^{(i)} - |v| + cv - |R|, \quad (18.4)$$

where the sum is empty for $m = 1$. Differentiation of the sum telescopes. Using $v^{(m)} = f$ and (18.1), the chain rules give

$$\mathcal{H}' = R'(f - \text{sign } R) = 0 \quad \text{almost everywhere.}$$

Off the residual zero set this is dual equality; on that set $R' = 0$ almost everywhere by the absolutely continuous level-set derivative property. For $m = 1$ the same displayed derivative follows directly. No sign function is differentiated.

All endpoint derivatives of v through order $m - 1$ vanish, so $\mathcal{H} = -|R|$ at both endpoints. Each summand in (18.4) has integral

$$\int_0^1 (-1)^{m-1-i} R^{(m-i)} v^{(i)} = - \int_0^1 (g - c)v = -J.$$

Indeed, integrate by parts i times; the clamped traces remove every boundary term, and $\int v = 0$. The remaining two contributions in (18.4) also integrate to $-J$. Hence

$$\mathcal{H} = -nJ = -A, \quad A = nJ > 0, \quad (18.5)$$

proving the endpoint identities before any restriction on c or either zero set.

18.2. The strict mean-parameter bound

We first record a zero lemma. If $h \in W^{m,\infty}$ has $h^{(m)}$ of one sign almost everywhere, then $m + 1$ distinct zeros force h to vanish throughout their convex hull. To prove this, take zeros $t_0 < \dots < t_m$. Their divided difference of order m is zero. The repeated fundamental theorem of calculus expresses this divided difference as the integral of $h^{(m)}$ over the simplex of convex combinations of these knots. Its induced density is positive almost everywhere throughout (t_0, t_m) . The one-signed derivative must therefore vanish almost everywhere there. Thus h is a polynomial of degree at most $m - 1$ on that interval, and its $m + 1$ zeros make it identically zero. The integral formula for $W^{m,\infty}$ follows by mollification from the smooth formula. In particular, a function of this kind with no zero interval has at most m distinct zeros.

Suppose now that $c \geq 1$. Then $R^{(m)}$ has one sign. If R has no zero interval, it has at most m zeros and $f = \text{sign } R$ almost everywhere. Choose a polynomial Q with roots exactly at the sign-changing zeros of R , and choose its overall sign to agree with R . Then $\deg Q \leq m$ and $fQ = |Q|$ almost everywhere. The positive integral of this product contradicts the moment constraints.

Otherwise take a maximal nontrivial zero interval $[\alpha, \beta]$ of R . Its derivative traces through order $m - 1$ vanish at both ends. Equation (18.1) gives $g = c$ almost everywhere inside it, so $c = 1$ and $v > 0$ almost everywhere there. The integral Taylor formulas give

$$R(t) = \frac{1}{(m-1)!} \int_t^\alpha (s-t)^{m-1} (g(s) - 1) ds \leq 0 \quad (t < \alpha)$$

and

$$R(t) = \frac{(-1)^m}{(m-1)!} \int_\beta^t (t-s)^{m-1} (g(s) - 1) ds \quad (t > \beta).$$

The signs are strict outside the maximal zero interval: equality at an outside point would force $g - 1 = 0$ almost everywhere between it and the interval, extending the zero interval by the same Taylor formula. Consequently $f = -1$ to the left and $f = (-1)^{m+1}$ to the right. The clamped endpoint traces imply

$$v(t) = -\frac{t^m}{m!} \quad (t < \alpha), \quad v(t) = -\frac{(1-t)^m}{m!} \quad (t > \beta). \quad (18.6)$$

Continuity against $v > 0$ almost everywhere inside the zero interval forces $\alpha = 0$ and $\beta = 1$. This would make R identically zero, contradicting $J > 0$. Thus $c < 1$. Applying the same argument to $-u$, $-R$, $-c$ gives $c > -1$.

18.3. Null zero sets and saturation

For a function in $W^{m,\infty}$, all derivatives through order m vanish almost everywhere on a level set. This follows by successively applying the absolutely continuous level-set derivative property, discarding a null set at each of the finitely many steps.

Suppose that $E = \{R = 0\}$ had positive measure. Equation (18.1) would give $g = c$ almost everywhere on E . Since g takes values only in $\{-1, 0, 1\}$ and $|c| < 1$, this forces $c = 0$ and $v = 0$ almost everywhere on E . Every term in (18.4) then vanishes almost everywhere on E , contradicting the nonzero constant (18.5). Hence E is null, proving (18.3).

If $\{v = 0\}$ had positive measure, the same level-set property would give $f = v^{(m)} = 0$ almost everywhere on it. This contradicts $|f| = 1$ almost everywhere. Thus the zero set of v is also null.

Finally, continuity and the nonzero endpoint values of R give a fixed nonzero sign on each endpoint neighborhood. Equation (18.3) makes f constant there. Integration of the clamped traces gives, on the respective neighborhoods,

$$v(t) = \text{sign } R(0) \frac{t^m}{m!}, \quad v(t) = (-1)^m \text{sign } R(1) \frac{(1-t)^m}{m!}. \quad (18.7)$$

This proves saturation and the endpoint assertions. We next prove finiteness and simplicity.

18.4. Finitely many simple zeros

An infinite set of residual zeros would have an accumulation point t_0 in the interior, because R is nonzero near both endpoints. Repeated Rolle arguments and $R \in C^{m-1}$ give

$$R^{(j)}(t_0) = 0 \quad (0 \leq j \leq m-1).$$

Equation (18.5) then gives $|v(t_0)| - cv(t_0) = A > 0$, so $v(t_0) \neq 0$. Its sign is fixed on a neighborhood. There (18.1) makes R a polynomial of degree exactly m , since $|c| < 1$. It cannot have infinitely many zeros on that neighborhood. Thus R has finitely many zeros.

The same argument applies to v . Its endpoint monomials exclude accumulation at the boundary. At an interior accumulation point, $v^{(j)}(t_0) = 0$ for $0 \leq j \leq m-1$, and (18.5) gives $|R(t_0)| = A > 0$. Hence f has one fixed sign nearby, making v a polynomial of degree exactly m , again a contradiction.

Both functions are therefore C^{m-1} splines of degree m : R has a knot only where v changes sign, while v has a knot only where R changes sign. All polynomial pieces have nonzero leading coefficient. Every zero has finite multiplicity at most m . Indeed, if all derivatives through $m-1$ vanish at a zero of either function, the conserved quantity makes the other function nonzero there; the first function then agrees locally with a polynomial of degree exactly m .

Let s be the number of sign changes of R , let r_0 be the number of sign changes of v , and let M be the sum of the multiplicities of all interior zeros of v . A C^{m-1} spline of degree m with k knots and nonzero leading coefficient on every piece has at most $m+k$ zeros counted with multiplicity. This follows by applying Rolle's theorem $m-1$ times: its $(m-1)$ st derivative is continuous and affine with nonzero slope on each of the $k+1$ pieces, so it has at most $k+1$ distinct zeros. Multiplicities are reduced by one at each differentiation, as in Section 8. Endpoint zeros are included when present.

Apply this bound first to v , whose endpoint zeros each have multiplicity m , and then to R , whose endpoint values are nonzero. If Z is the sum of the multiplicities of the interior residual zeros, then

$$2m + M \leq m + s, \quad s \leq Z \leq m + r_0 \leq m + M. \quad (18.8)$$

Thus equality holds throughout. In particular $M = r_0$: every zero of v changes sign and has multiplicity one. Also $Z = s = m + r_0$, so every residual zero is simple and changes sign. Writing $r = r_0$, there are exactly $n + r - 1$ residual zeros. Since $\int v = 0$ and $v \neq 0$, one has $r \geq 1$. For $m = 1$ the same count applies directly to continuous piecewise affine functions. This proves Theorem 19. $\square$

Corollary (uniform critical-point bound over all exponents). For each fixed $n \geq 3$, there is a finite bound on the number of critical points of any variation maximizer, uniform over $1 < p \leq \infty$. The bound is not made explicit.

Proof. Theorem 11 bounds the count by one when p is sufficiently close to one, and Theorem 12 gives uniform bounds on compact subintervals of $(1, \infty)$. It remains to treat sequences $p_j \rightarrow \infty$, allowing $p_j = \infty$.

The constants $V_{n,p}$ decrease with p and are bounded below by $V_{n,\infty}$. For normalized maximizers at finite p_j , the highest derivative has L^s norm at most one once $p_j \geq s$, for every fixed finite s . Compactness of its successive integrals, weak compactness for each $s > 1$, and a diagonal extraction give a limit whose highest derivative has every finite L^s norm at most one and hence L^∞ norm at most one. The first derivatives converge uniformly. Their variations converge to the limit of the constants, proving $V_{n,p} \rightarrow V_{n,\infty}$ and showing that every such limit maximizes endpoint variation. The same compactness holds for sequences already at the endpoint.

Choose best polynomial residuals for the sequence. Their coefficients are bounded: $\|P_j\|_1 \leq 2\|H_{g_j}\|_\infty$, using the best-approximation inequality and finite-dimensional norm equivalence. By the just-proved regularity of the endpoint limit, its first derivative has only finitely many zeros, so uniform convergence gives $g_j \rightarrow g$ almost everywhere and in L^1 . Thus, after extracting polynomial coefficients, $R_j \rightarrow R$ uniformly. For finite exponents, $q_j \rightarrow 1$ and comparison against every fixed polynomial shows that R is a best L^1 residual for the endpoint load. The same assertion follows directly for endpoint residuals.

Theorem 19 gives nonzero endpoint values of R . Uniform residual convergence therefore fixes the highest-derivative signs on two common endpoint neighborhoods, by (6.6) for finite exponents and (18.3) at the endpoint. Successive integration from the clamped traces excludes critical points there. On the interior, $u'_j \rightarrow u'$ also holds in C^1 : the functions u''_j have uniform Holder control from any fixed L^s bound on the highest derivative with $s > 1$, also when $n = 3$. The endpoint limit's critical points are all simple. On disjoint small neighborhoods of those points, the second derivative has one fixed nonzero sign, and on the remaining compact set the first derivative stays away from zero. The sequence therefore has exactly the same critical-point count for all sufficiently large indices. This excludes an unbounded sequence of counts as $p \rightarrow \infty$ and completes the proof. $\square$

18.5. A local obstruction to the order-three residual bound

For $n = 3$, the mixed term in (18.4) is $R'v'$, which vanishes at a residual extremum. For $n = 4$, it is $-R''v' + R'v''$. At $R' = 0$ the first term survives and has no prescribed sign. The following exact local solutions show why the conserved quantity alone cannot supply the order-three bound.

On $[-1/4, 1/4]$, put $c = 0$ and

$$R(t) = 2 - \frac{t^2}{2} - \frac{t^3}{6}, \quad v(t) = 1 + 2t + \frac{t^3}{6}. \quad (18.9)$$

Here $R > 1$ and $v > 0$, while $R''' = -\text{sign } v$, $v''' = \text{sign } R$ and $\mathcal{H} = -1$. At zero, R has a strict local maximum of value two, exceeding the formal level $A = 1$ obtained from $-\mathcal{H}$. This is a local stationary solution; it is not clamped and is not a global maximizer.

The same obstruction is present at finite exponent. For $q = 2$, keep this R and take

$$v(t) = 1 + \frac{5}{2}t + \frac{t^3}{3} - \frac{t^5}{120} - \frac{t^6}{720}. \quad (18.10)$$

Then $v''' = R$, $R''' = -\text{sign } v$, and $-R''v' + R'v'' - |v| - R^2/2 = -1/2$. Again the formal level is $A = 1$ while $R(0) = 2$. Direct triangle bounds on the stated interval give $R \geq 755/384 > 1$ and $v \geq 218107/589824 > 0$; in (18.9), $v \geq 191/384 > 0$.

These examples disprove only an inference from the local equations and negative conserved level to the residual bound. They do not disprove a bound with global maximizing hypotheses, or the variation

conjecture. The remaining problem is to control the additional derivative traces and construct a better admissible competitor for every multiple-peak maximizer.

19. Endpoint second variation and algebraic sharp constants

The finite simple residual zeros in Theorem 19 permit a second-variation calculation at the uniform endpoint. The second derivative is a finite weighted sum over those zeros. It does not vanish when the conjugate exponent tends to one.

Theorem 20 (endpoint switching Hessian). Let $n \geq 3$ and let u be a normalized variation maximizer in $W_0^{n,\infty}(0,1)$. Write $\tau_1 < \dots < \tau_r$ for its critical points and σ_i for the sign of u' immediately to the left of τ_i . Let R be its best polynomial residual, with simple zeros $\xi_1 < \dots < \xi_s$, where $s = n + r - 1$. Define

$$\mu = 2 \sum_{k=1}^s \frac{\delta_{\xi_k}}{|R'(\xi_k)|}, \quad L_i(t) = \frac{(\tau_i - t)_+^{n-2}}{(n-2)!}, \quad B_i = L_i - \Pi_\mu L_i, \quad (19.1)$$

where Π_μ is the orthogonal projection onto the restrictions of $\mathcal{P}_{n-1}$ to the residual zeros. The nearby switching objective

$$J(\tau) = \min_{P \in \mathcal{P}_{n-1}} \int_0^1 \left| 2 \sum_{i=1}^r \sigma_i K_{\tau_i} - P \right|, \quad K_a(t) = \frac{(a-t)_+^{n-1}}{(n-1)!}, \quad (19.2)$$

is twice continuously differentiable, and at this configuration

$$J_{ij} = 4\sigma_i\sigma_j \int B_i B_j d\mu + 2\sigma_i\delta_{ij}u''(\tau_i). \quad (19.3)$$

Consequently, with $D_i = -\sigma_i u''(\tau_i) > 0$ and $G_{ij} = \sigma_i\sigma_j \int B_i B_j d\mu$, one has

$$2G \preceq \text{diag}(D_1, \dots, D_r), \quad \text{rank } G \leq r - 1. \quad (19.4)$$

19.1. Differentiating the moving residual zeros

The Gram matrix of $\mathcal{P}_{n-1}$ for μ is positive definite: there are $s \geq n$ distinct nodes, and a nonzero polynomial of degree at most $n-1$ cannot vanish at all of them. For a perturbation h of a residual with simple zeros, differentiation of the signed moments gives

$$D \left(\int_0^1 \text{sign } R Q \right) [h] = 2 \sum_{k=1}^s \frac{h(\xi_k) Q(\xi_k)}{|R'(\xi_k)|} = \int h Q d\mu. \quad (19.5)$$

Indeed, a residual zero moves with velocity $-h(\xi_k)/R'(\xi_k)$, and the sign jump there is $2 \text{sign } R'(\xi_k)$. Differentiating the endpoints of the finitely many sign intervals gives (19.5), including its positive sign and factor two. The same formula holds for a continuous test kernel in place of Q .

For $n \geq 3$, $K_a(t)$ is jointly continuously differentiable in a, t . Thus the simple-root implicit function theorem applies even when a residual zero equals a load switch. It provides continuously differentiable root locations as functions of the switch positions and polynomial coefficients. All nearby residuals retain exactly these simple zeros: their derivatives have fixed nonzero signs on small root neighborhoods, and the residual is bounded away from zero on the remaining compact set, including both endpoints.

The normal equations are $\int \text{sign } R Q = 0$ for $Q \in \mathcal{P}_{n-1}$. Their derivative in the polynomial coefficients is the negative Gram matrix for μ . The implicit function theorem therefore gives a continuously differentiable polynomial branch satisfying these equations. The zero signed moments and convexity of the L^1 objective make it a branch of global best approximants.

Each best polynomial on this branch is unique. If two polynomials were best, equality in convexity would force their residuals to have the same sign almost everywhere. At every simple zero of the first residual, continuity forces the second to vanish too; otherwise their signs disagree on one side. Their difference is a polynomial of degree at most $n - 1$ with at least n distinct zeros and hence is zero.

Differentiating the normal equations and the objective now gives

$$\frac{\partial R}{\partial \tau_i} = 2\sigma_i B_i, \quad J_i = 2\sigma_i U'(\tau_i), \quad U = I^n(\text{sign } R). \quad (19.6)$$

At the maximizing configuration, Theorem 19 gives $U = u$. Differentiate the integral defining $U'(\tau_i)$ using (19.5) and the moving evaluation point. Polynomial orthogonality replaces L_i by B_i in the residual contribution and yields (19.3). These finite-root formulas also show that the gradient is continuously differentiable. When $n = 3$, the derivative of the moving L_i kernel is an indicator; integration against the bounded sign control justifies the calculation by dominated convergence. No ordinary derivative of the discontinuous sign function is asserted.

Every nearby sign load in (19.2) has dual norm at most $V_{n,\infty}$, with equality at the given maximizer. Its Hessian is therefore negative semidefinite, giving the first assertion in (19.4). The orthogonal complement of polynomial data in the s -dimensional node space has dimension $s - n = r - 1$, giving the rank bound. $\square$

The same local differentiation proof applies to any configuration with at least n simple residual zeros and nonzero endpoint residuals, before a maximality assertion. At a maximizer the rank bound supplies a strictly negative Hessian direction; for $r = 1$ the Hessian is exactly $-2D_1 < 0$. A negative direction is compatible with a local maximum. Excluding higher counts still requires a positive direction or a global value comparison. In particular, the finite-exponent exclusion proof cannot be transferred by formally setting $q = 1$: the measure in (19.1) replaces the vanishing factor in that formal substitution.

19.2. Semialgebraic description at a fixed control count

Theorem 21 (algebraicity at the uniform endpoint). For every integer $n \geq 2$, the sharp constant $V_{n,\infty}$ is a real algebraic number. There exists a normalized variation maximizer whose control knots and polynomial coefficients are all real algebraic numbers.

Proof. Fix an integer $s \geq n$. For ordered knots $0 = x_0 < x_1 < \cdots < x_s < x_{s+1} = 1$, take the alternating control $f = (-1)^j$ on (x_j, x_{j+1}) . It satisfies the required polynomial moment conditions precisely when

$$\sum_{j=0}^s (-1)^j (x_{j+1}^{k+1} - x_j^{k+1}) = 0, \quad 0 \leq k < n. \quad (19.7)$$

Its integrated potential is

$$u_x(t) = \frac{1}{n!} \left[t^n + 2 \sum_{j=1}^s (-1)^j (t - x_j)_+^n \right]. \quad (19.8)$$

The left clamped traces vanish by construction, and (19.7) is equivalent to all right clamped traces. The highest derivative has norm one.

We claim that the graph of the total variation $T_s(x)$ is semialgebraic over $\mathbb{Q}$, meaning that it is described by a finite Boolean combination of polynomial equalities and inequalities with rational coefficients. On each control interval, (19.8) is a polynomial in t with rational polynomial coefficients in the knots. Its derivative has degree $n - 1$ with nonzero leading coefficient. List all its interior critical points in increasing order, using at most $n - 1$ variables per interval. Polynomial conditions require that these points are zeros; a universal real quantifier requires that there are no other zeros in the open interval. There are finitely many choices of list lengths.

Adjoin the two interval endpoints to each list. The variation on that interval is the sum of the absolute differences between consecutive polynomial values. Absolute value is expressed by polynomial sign alternatives. Summing these expressions gives a first-order formula over $\mathbb{Q}$ for $z = T_s(x)$. Quantifier elimination over real closed fields [B, Theorem 2.1] makes this graph semialgebraic over $\mathbb{Q}$. This argument uses the finite critical-point description, not a claim that arbitrary integrals of semialgebraic functions are semialgebraic.

The set of attained values under (19.7) and the strict knot inequalities is consequently a one-dimensional semialgebraic set over $\mathbb{Q}$. Any attained finite maximum of such a set is real algebraic: in a quantifier-free description, signs of all nonzero rational polynomials are constant near any transcendental point, so such a point cannot be a finite endpoint of the set.

Theorem 19 supplies a global maximizer with a finite control count s . Change its overall sign to make its first control positive. It then has representation (19.8), while every admissible configuration at this same s has variation at most $V_{n,\infty}$. The set of attained values at this s therefore has attained maximum exactly $V_{n,\infty}$, proving algebraicity.

Finally impose $z = V_{n,\infty}$ in the first-order formula. It has a solution over $\mathbb{R}$. Quantifier elimination transfers this statement, with its real algebraic coefficients, to the real closed field of real algebraic numbers; this field is included in the notation of [B]. The resulting knots and auxiliary critical points are algebraic. Formula (19.8) gives a maximizing spline with algebraic coefficients on all pieces. $\square$

This proof does not assert algebraicity of every maximizing knot configuration, an explicit algebraic degree bound, or an effective stopping count for searching over s . The uniform count bound in Section 18 is implicit. Theorem 21 determines the arithmetic nature of the sharp constant without computing it or settling the higher-order variation identity.

20. Remaining questions

The order-three case of [NS, Conjecture 4.14] is now complete at every exponent: Theorem 17 gives the finite-exponent identity and equality classification, Theorem 18 gives the complete uniform endpoint, and Theorem 9 supplies the measure endpoint. The full all-order conjecture remains unresolved. In every order, Theorems 5 and 11 establish exponent intervals around two and immediately above one; their widths and overlap are not known. Theorem 10 proves height symmetry and uniqueness at every finite exponent.

For orders $n \geq 4$, Theorem 12 supplies finite nondegenerate critical points and simple dual-residual zeros. Theorem 13 excludes every two-switch maximizer, and Theorem 14 excludes midpoint-symmetric three-switch maximizers and restricts every remaining three-switch candidate. Excluding nonsymmetric three-switch configurations and all higher counts remains the finite-exponent task in those orders. At the uniform endpoint, Theorem 19 proves saturation, finite simple critical points and exactly $n + r - 1$ simple residual zeros in every order. The critical-point count is uniformly bounded over all $1 < p \leq \infty$

for each fixed order, but no explicit bound or higher-order single-peak conclusion is proved. The positive action in Sections 16 and 17 uses the quadratic residual pieces specific to order three. Section 18 identifies the additional mixed derivative terms and gives exact local obstructions to reusing the same pointwise argument; no higher-order action comparison is proved here.

Theorem 20 provides the endpoint switching Hessian, including its discrete root weights and polynomial projection. Determining a positive Hessian direction for all multiple-peak candidates remains open. Theorem 21 proves that every uniform-endpoint sharp variation constant is algebraic and admits an algebraic maximizing spline. It gives neither an explicit degree bound nor an effective stopping count for a search over control knots.

Theorem 15 rules out a reduction using the same identity for arbitrary derivative weights. The special weights generated by an actual maximizer remain a possible restricted route in higher orders. The stronger centroid and knot-reflection inequalities in the research notes remain unproved and are not needed for the results above.

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